# COURSE CODE
SBT-DF203

# COURSE TITLE
Basic Networking Skills for Digital Forensics

# LAB TITLE
WEP40 Wireless Packet Decryption and Aircrack Forensics

# NAME
Kumpon Emmanuel Nandom

# STUDENT ID
2025/FWSD/11325

# DATE
25/09/2026

# PLATFORM USED
Kali Linux

# *INSTRUCTOR*
Mr. Aminu Idris, AMCPN
Commandant

# Executive Summary

This report describes the steps taken to prepare evidence and analyze forensics for the CodeGate CTF 2015 “Good Crypto” WEP40 capture used in SBT-DF203 Lab 9. All work was done offline in a separate Kali Linux virtual machine using only the provided training file. At no point did we connect to or capture data from a live wireless network, which follows the lab's rules on legality, ethics, and safety.

The report includes the lab brief in detail: the structure of the lab folder and evidence preparation, inventory of captures, identification of WEP protection, recovery of the WEP40 key, offline decryption, analysis of endpoints and conversations, extraction of objects, the optional passphrase challenge (which was not performed), WEP security analysis, and the completed findings worksheet.

Mini Evidence and Chain-of-Custody Worksheet

| | |
|---|---|
| **Case/lab identifier** | SBT-DF203-Lab9-Kumpon_Emmanuel_Nandom |
| **Trainee name** | Kumpon Emmanuel Nandom |
| **Date and time started** | 25 September 2026, 19:56 WAT |
| **Evidence file name(s)** | evidence/file.xz (original); working/file_working. xz and working/file_working (working copies); working/file_working-dec (decrypted output) |
| **Source or generation method** | Downloaded through 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz' |
| **Original SHA-256** | dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz |
| **Working-copy SHA-256** | dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working. xz<br>c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working |
| **Analysis workstation/VM** | Kali Linux VM, local user ponzhi@kali |
| **Notes on any changes** | We kept the original evidence unchanged and stored it under "evidence." We performed our analysis on a working copy that was uncompressed, found in "working/," and later on a decrypted file located in "working/file_working-dec." We recorded all hashes before and after each step of the process, and we found no discrepancies. |

# Lab Folder Structure and Evidence Preparation

Before any evidence was downloaded or created, we set up a standard folder structure to organize the evidence. This structure separates the original evidence, working copies, and exported objects.

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print
```

```
┌──(ponzhi㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}

┌──(ponzhi㉿kali)-[~]
└─$ cd ~/SBT-DF203-Lab9

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ pwd
/home/ponzhi/SBT-DF203-Lab9

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ find . -maxdepth 1 -type d -print
.
./evidence
./working
./exported
./reports
./scripts
./screenshots
```

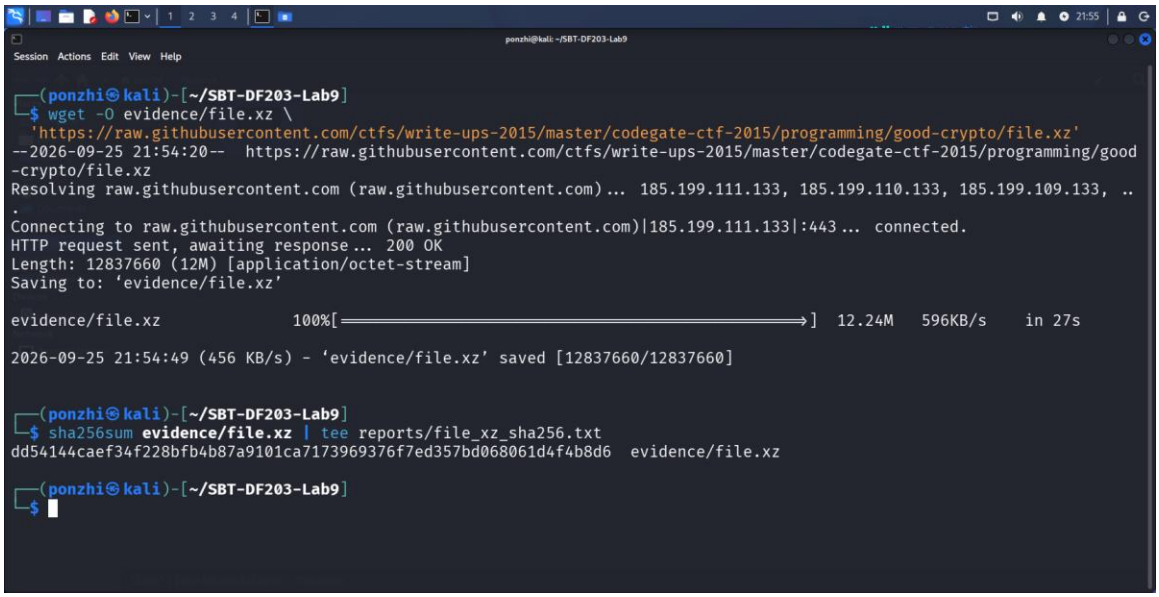
Required tools were confirmed installed (aircrack-ng, Wireshark/tshark, foremost, binwalk, xz-utils)

```
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
```

```
┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Fetched 3,617 kB in 19s (190 kB/s)
471 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
aircrack-ng is already the newest version (1:1.7+git20230807.4bf83f1a-3).
aircrack-ng set to manually installed.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
binwalk is already the newest version (2.4.3+dfsg1-3).
binwalk set to manually installed.
xz-utils is already the newest version (5.8.3-1).
The following packages were automatically installed and are no longer required:
  bluez-hcidump              libbluray3       libopenjph0.27     libvidstab1.1      python3.13-dev
  firmware-marvell-prestera  libhdf5-310      libperl5.40        libx265-215        python3.13-minimal
  iucode-tool                libhdf5-hl-310   libpoppler156      perl-modules-5.40  python3.13-tk
  libabsl20260107            libntfs-3g89t64  libpython3.13-dev  python3.13         python3.13-venv
Use 'sudo apt autoremove' to remove them.

Installing:
  foremost
```

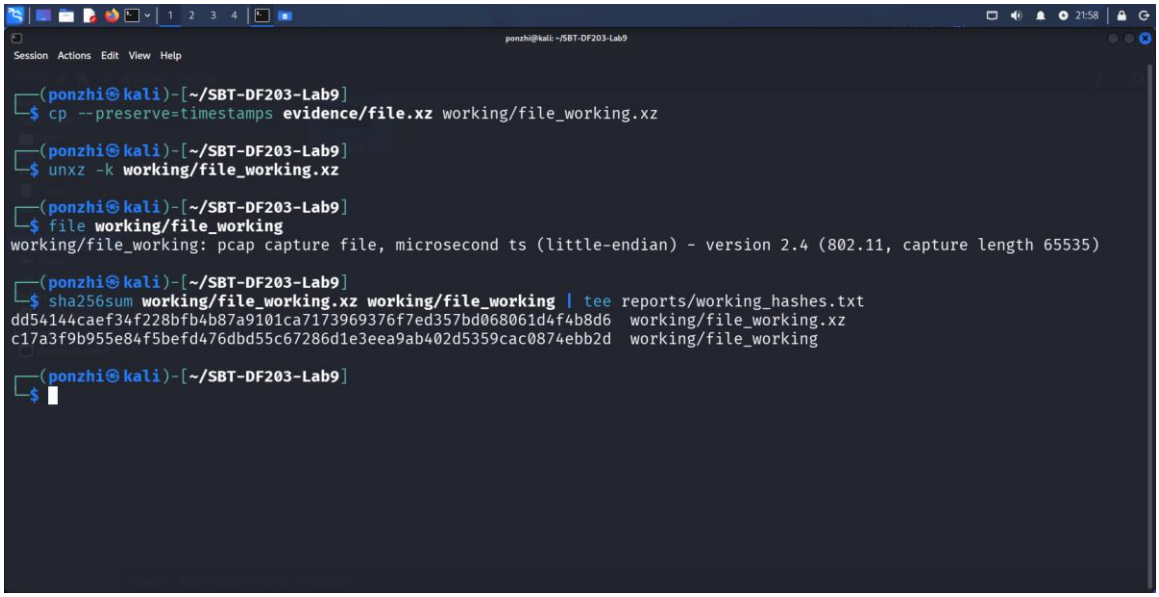
The historical evidence file was retrieved over HTTPS and hashed immediately upon download, before any local modification.

```
wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
```




A working copy was created, decompressed, identified by file type, and rehashed to confirm evidence integrity was preserved across the copy/decompress steps.

```
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

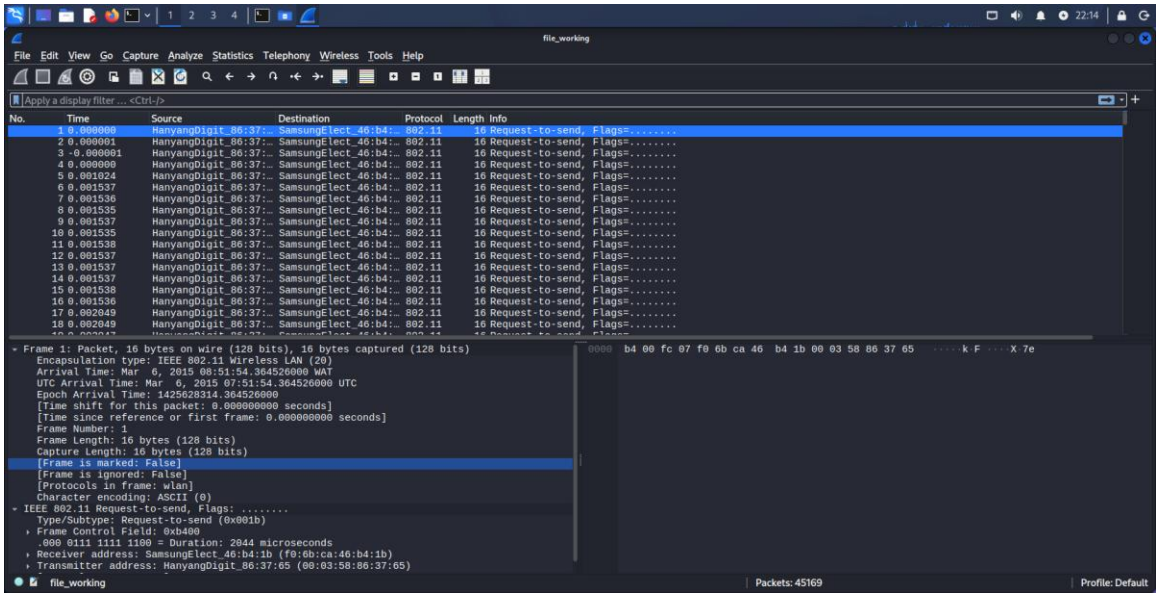



## Evidence hashes recorded in Section 6

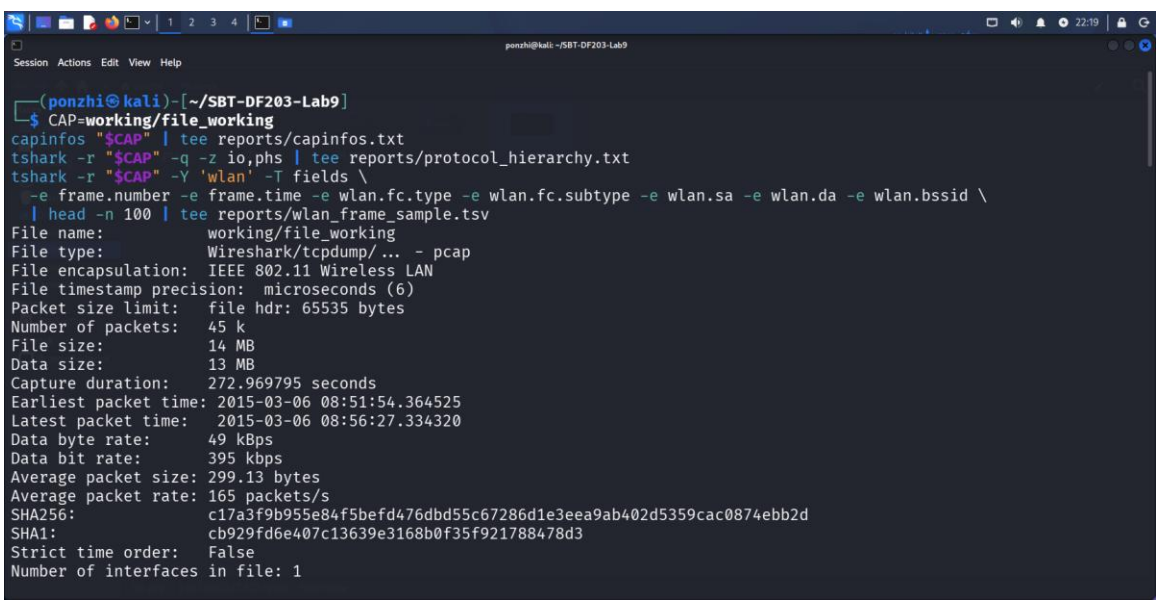
| File | SHA-256 | Notes |
|---|---|---|
| evidence/file.xz | dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 | Original, as downloaded |
| working/file_working.xz | dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 | Copy of original — hash matches |
| working/file_working | c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d | Decompressed pcap, ready for analysis |

# Part A: Inventory the Wireless Capture

The decompressed capture was opened in Wireshark to confirm it parses as a valid 802.11 capture and to visually inspect frame structure before running command-line tooling.



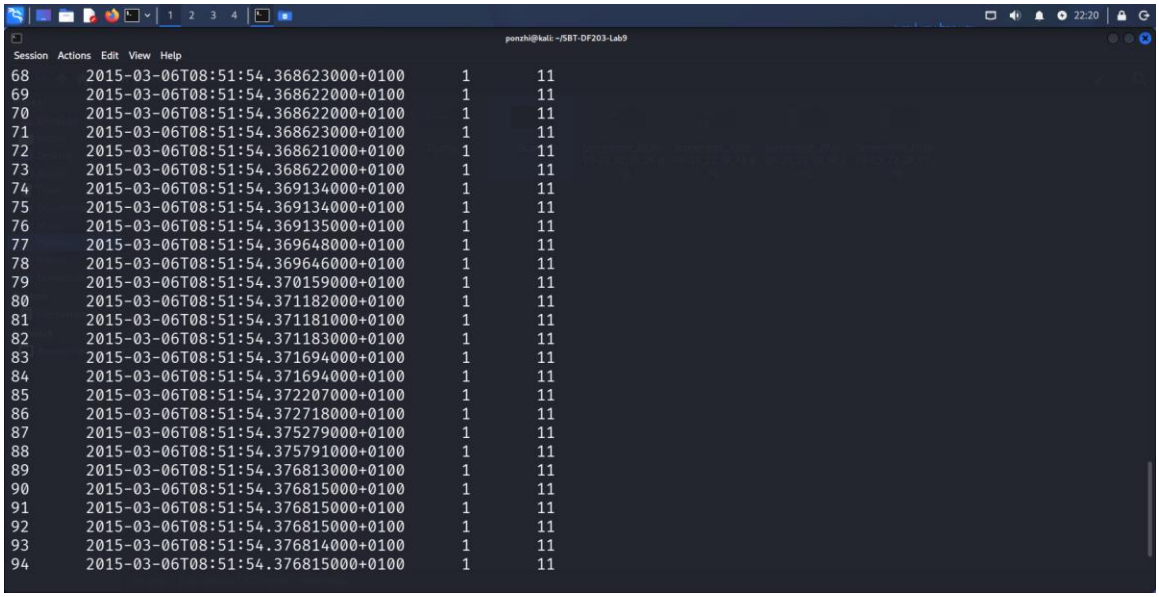

```
CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv
```



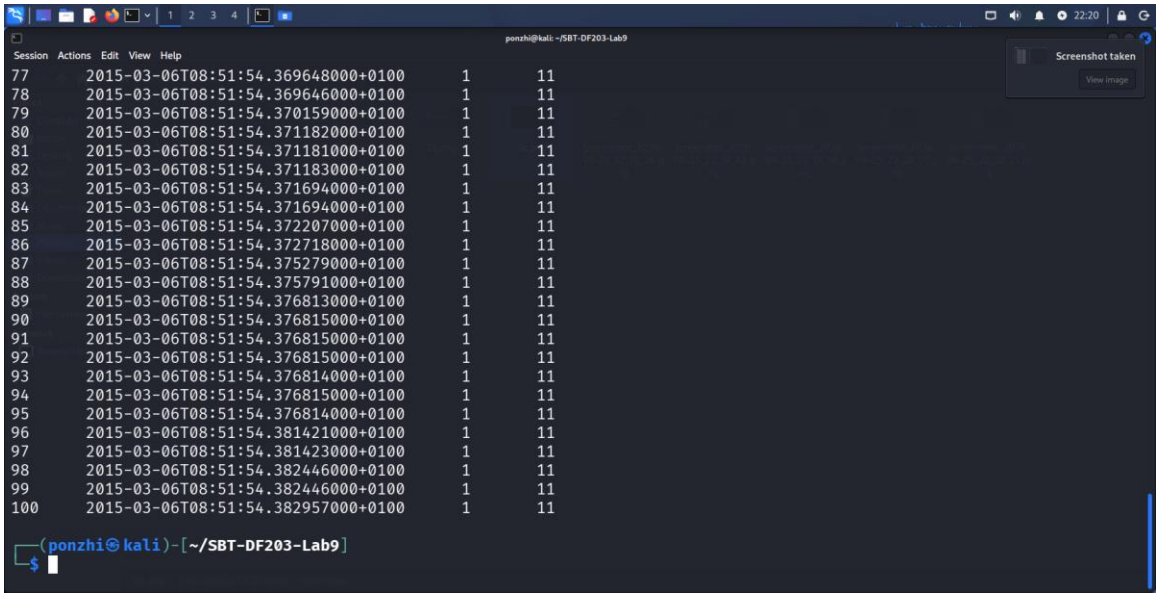
```
Number of interfaces in file: 1
Interface #0 info:
                    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
                    Capture length = 65535
                    Time precision = microseconds (6)
                    Time ticks per second = 1000000
                    Number of stat entries = 0
                    Number of packets = 45169

===================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:45169 bytes:13511274
  wlan                                   frames:45169 bytes:13511274
    data                                 frames:15713 bytes:13104128
===================================================================
1       2015-03-06T08:51:54.364526000+0100      1       11
2       2015-03-06T08:51:54.364527000+0100      1       11
3       2015-03-06T08:51:54.364525000+0100      1       11
4       2015-03-06T08:51:54.364526000+0100      1       11
5       2015-03-06T08:51:54.365550000+0100      1       11
6       2015-03-06T08:51:54.366063000+0100      1       11
7       2015-03-06T08:51:54.366062000+0100      1       11
8       2015-03-06T08:51:54.366061000+0100      1       11
9       2015-03-06T08:51:54.366063000+0100      1       11
10      2015-03-06T08:51:54.366061000+0100      1       11
```

```
10      2015-03-06T08:51:54.366061000+0100      1       11
11      2015-03-06T08:51:54.366064000+0100      1       11
12      2015-03-06T08:51:54.366063000+0100      1       11
13      2015-03-06T08:51:54.366063000+0100      1       11
14      2015-03-06T08:51:54.366063000+0100      1       11
15      2015-03-06T08:51:54.366064000+0100      1       11
16      2015-03-06T08:51:54.366062000+0100      1       11
17      2015-03-06T08:51:54.366575000+0100      1       11
18      2015-03-06T08:51:54.366575000+0100      1       11
19      2015-03-06T08:51:54.366573000+0100      1       11
20      2015-03-06T08:51:54.366574000+0100      1       11
21      2015-03-06T08:51:54.366575000+0100      1       11
22      2015-03-06T08:51:54.366575000+0100      1       11
23      2015-03-06T08:51:54.366575000+0100      1       11
24      2015-03-06T08:51:54.366574000+0100      1       11
25      2015-03-06T08:51:54.366574000+0100      1       11
26      2015-03-06T08:51:54.366574000+0100      1       11
27      2015-03-06T08:51:54.366574000+0100      1       11
28      2015-03-06T08:51:54.366574000+0100      1       11
29      2015-03-06T08:51:54.366574000+0100      1       11
30      2015-03-06T08:51:54.366575000+0100      1       11
31      2015-03-06T08:51:54.366573000+0100      1       11
32      2015-03-06T08:51:54.366575000+0100      1       11
33      2015-03-06T08:51:54.366574000+0100      1       11
34      2015-03-06T08:51:54.366575000+0100      1       11
35      2015-03-06T08:51:54.366575000+0100      1       11
36      2015-03-06T08:51:54.366574000+0100      1       11
```

```
42      2015-03-06T08:51:54.366573000+0100      1       11
43      2015-03-06T08:51:54.366574000+0100      1       11
44      2015-03-06T08:51:54.367085000+0100      1       11
45      2015-03-06T08:51:54.367086000+0100      1       11
46      2015-03-06T08:51:54.367087000+0100      1       11
47      2015-03-06T08:51:54.367089000+0100      1       13
48      2015-03-06T08:51:54.367085000+0100      1       11
49      2015-03-06T08:51:54.367087000+0100      1       11
50      2015-03-06T08:51:54.367087000+0100      1       11
51      2015-03-06T08:51:54.367087000+0100      1       11
52      2015-03-06T08:51:54.367087000+0100      1       11
53      2015-03-06T08:51:54.367086000+0100      1       11
54      2015-03-06T08:51:54.367633000+0100      0       8       00:26:66:55:97:d6       ff:ff:ff:ff:ff:ff       00:26:66
:55:97:d6
55      2015-03-06T08:51:54.368111000+0100      1       11
56      2015-03-06T08:51:54.368111000+0100      1       11
57      2015-03-06T08:51:54.368110000+0100      1       11
58      2015-03-06T08:51:54.368110000+0100      1       11
59      2015-03-06T08:51:54.368110000+0100      1       11
60      2015-03-06T08:51:54.368110000+0100      1       11
61      2015-03-06T08:51:54.368622000+0100      1       11
62      2015-03-06T08:51:54.368623000+0100      1       11
63      2015-03-06T08:51:54.368621000+0100      1       11
64      2015-03-06T08:51:54.368622000+0100      1       11
65      2015-03-06T08:51:54.368621000+0100      1       11
66      2015-03-06T08:51:54.368623000+0100      1       11
67      2015-03-06T08:51:54.368622000+0100      1       11
```



```
68      2015-03-06T08:51:54.368623000+0100      1       11
69      2015-03-06T08:51:54.368622000+0100      1       11
70      2015-03-06T08:51:54.368622000+0100      1       11
71      2015-03-06T08:51:54.368623000+0100      1       11
72      2015-03-06T08:51:54.368621000+0100      1       11
73      2015-03-06T08:51:54.368622000+0100      1       11
74      2015-03-06T08:51:54.369134000+0100      1       11
75      2015-03-06T08:51:54.369134000+0100      1       11
76      2015-03-06T08:51:54.369135000+0100      1       11
77      2015-03-06T08:51:54.369648000+0100      1       11
78      2015-03-06T08:51:54.369646000+0100      1       11
79      2015-03-06T08:51:54.370159000+0100      1       11
80      2015-03-06T08:51:54.371182000+0100      1       11
81      2015-03-06T08:51:54.371181000+0100      1       11
82      2015-03-06T08:51:54.371183000+0100      1       11
83      2015-03-06T08:51:54.371694000+0100      1       11
84      2015-03-06T08:51:54.371694000+0100      1       11
85      2015-03-06T08:51:54.372207000+0100      1       11
86      2015-03-06T08:51:54.372718000+0100      1       11
87      2015-03-06T08:51:54.375279000+0100      1       11
88      2015-03-06T08:51:54.375791000+0100      1       11
89      2015-03-06T08:51:54.376813000+0100      1       11
90      2015-03-06T08:51:54.376815000+0100      1       11
91      2015-03-06T08:51:54.376815000+0100      1       11
92      2015-03-06T08:51:54.376815000+0100      1       11
93      2015-03-06T08:51:54.376814000+0100      1       11
94      2015-03-06T08:51:54.376815000+0100      1       11
```



```
77      2015-03-06T08:51:54.369648000+0100      1       11
78      2015-03-06T08:51:54.369646000+0100      1       11
79      2015-03-06T08:51:54.370159000+0100      1       11
80      2015-03-06T08:51:54.371182000+0100      1       11
81      2015-03-06T08:51:54.371181000+0100      1       11
82      2015-03-06T08:51:54.371183000+0100      1       11
83      2015-03-06T08:51:54.371694000+0100      1       11
84      2015-03-06T08:51:54.371694000+0100      1       11
85      2015-03-06T08:51:54.372207000+0100      1       11
86      2015-03-06T08:51:54.372718000+0100      1       11
87      2015-03-06T08:51:54.375279000+0100      1       11
88      2015-03-06T08:51:54.375791000+0100      1       11
89      2015-03-06T08:51:54.376813000+0100      1       11
90      2015-03-06T08:51:54.376815000+0100      1       11
91      2015-03-06T08:51:54.376815000+0100      1       11
92      2015-03-06T08:51:54.376815000+0100      1       11
93      2015-03-06T08:51:54.376814000+0100      1       11
94      2015-03-06T08:51:54.376815000+0100      1       11
95      2015-03-06T08:51:54.376814000+0100      1       11
96      2015-03-06T08:51:54.381421000+0100      1       11
97      2015-03-06T08:51:54.381423000+0100      1       11
98      2015-03-06T08:51:54.382446000+0100      1       11
99      2015-03-06T08:51:54.382446000+0100      1       11
100     2015-03-06T08:51:54.382957000+0100      1       11

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$
```

## Capture inventory summary

| | |
|---|---|
| **File name** | working/file_working |
| **File type** | Wireshark/tcpdump/... pcap, IEEE 802.11 Wireless LAN, microsecond precision |
| **Number of packets** | 45,169 (frame: 45,169 frames / 13,511,274 bytes; wlan: 45,169 frames / 13,511,274 bytes; data: 15,713 frames / 13,104,128 bytes) |
| **File / data size** | 14 MB / 13 MB |
| **Capture duration** | 272.969795 seconds (2015-03-06 08:51:54.364525 → 08:56:27.334320) |
| **Average rate** | 165 packets, 395 kbps |
| **SHA-256 / SHA-1 (capinfos)** | c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d |

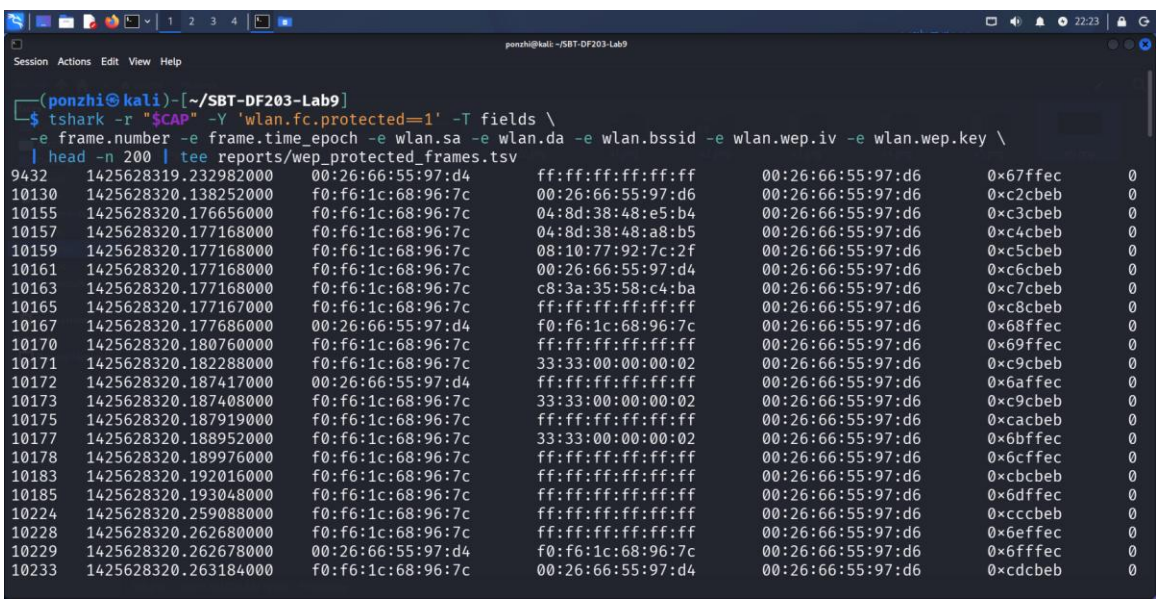
### 802.11 Frame Categories Observed

| Frame Category | Type Value | Purpose | Example Evidence |
|---|---|---|---|
| Management | 0 | Association, authentication, beacons | Frame 54: Beacon SA/BSSID 00:26:66:55:97:d6 → DA ff:ff:ff:ff:ff:ff |
| Control | 1 | Flow control and acknowledgements | Frame 1: Request-to-Send Transmitter 00:03:58:86:37:65 → Receiver f0:6b:ca:46:b4:1b |
| Data | 2 | Carries network payload | Frame 10130 (WEP-protected): SA f0:f6:1c:68:96:7c → BSSID 00:26:66:55:97:d6, IV 0xc2cbeb |

## Part B: Identify WEP Protection and Key Parameters

Protected (WEP) data frames were isolated with a display filter, and the IV field was extracted and tallied to demonstrate IV reuse the core weakness that makes WEP40 breakable.

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv
```




```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

```
(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
      7 0x42e8eb
      7 0x28d3eb
      6 0x95dfeb
      5 0x9415ed
      5 0x86e5eb
      5 0x7ae5eb
      5 0x28e7eb
      5 0x19e5eb
      4 0xd5d7eb
      4 0xd3e8eb
      4 0xc915ed
      4 0xc7d8eb
      4 0xade4eb
      4 0x73e5eb
      4 0x65d8eb
      4 0x4ad5eb
      4 0x31e7eb
      4 0x2723ed
      4 0x10e7eb
      3 0xffe5eb
      3 0xf609ed
      3 0xa307ed
      3 0x97dbeb
```

```
      5 0x28e7eb
      5 0x19e5eb
      4 0xd5d7eb
      4 0xd3e8eb
      4 0xc915ed
      4 0xc7d8eb
      4 0xade4eb
      4 0x73e5eb
      4 0x65d8eb
      4 0x4ad5eb
      4 0x31e7eb
      4 0x2723ed
      4 0x10e7eb
      3 0xffe5eb
      3 0xf609ed
      3 0xa307ed
      3 0x97dbeb
      3 0x96d1eb
      3 0x92e8eb
      3 0x91e8eb
      3 0x8adeeb
      3 0x630aed
      3 0x56e8eb
      3 0x3c1aed

(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
$
```

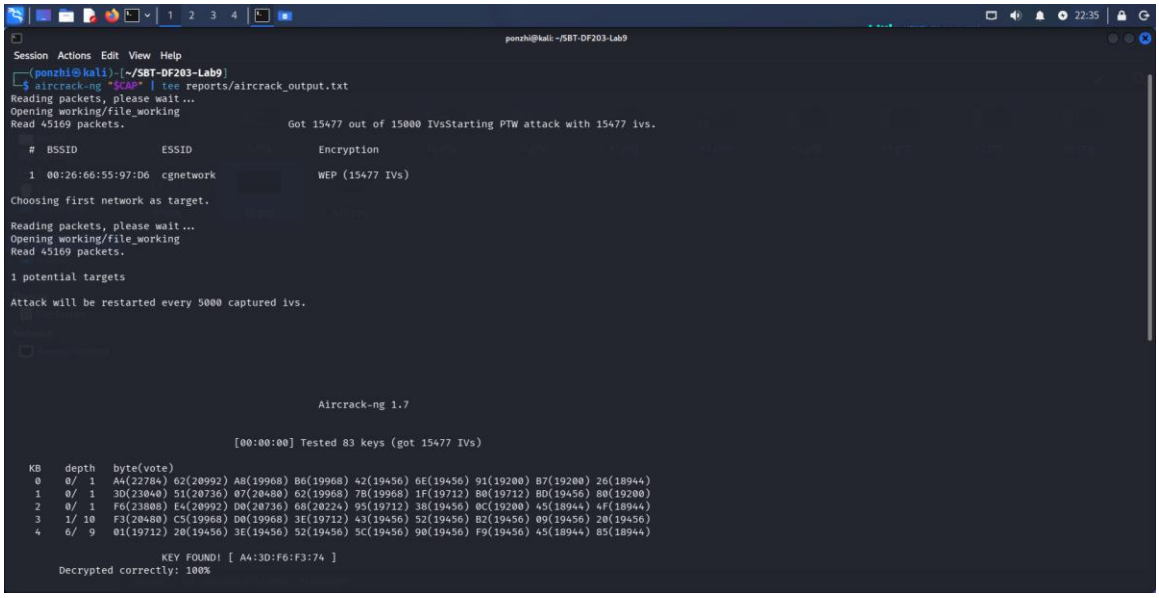
## WEP identification summary

| | |
|---|---|
| **BSSID** | 00:26:66:55:97:d6 |
| **Protected frame count** | 15,477 WEP data frames (of 45,169 total packets) |
| **Repeated IV evidence** | Top IVs repeat 4–7 times each within the capture; clear IV reuse |
| **WEP40 key structure** | 64-bit total keystream input: 24-bit IV + 40-bit secret key |
| **Why this is unsafe** | The 24-bit IV space (16.7M values) is exhausted quickly on an active network, forcing IV reuse; reused IVs produce identical RC4 keystreams, letting an attacker XOR ciphertexts to recover plaintext relationships and, via Aircrack-ng's statistical attacks, the secret key itself |

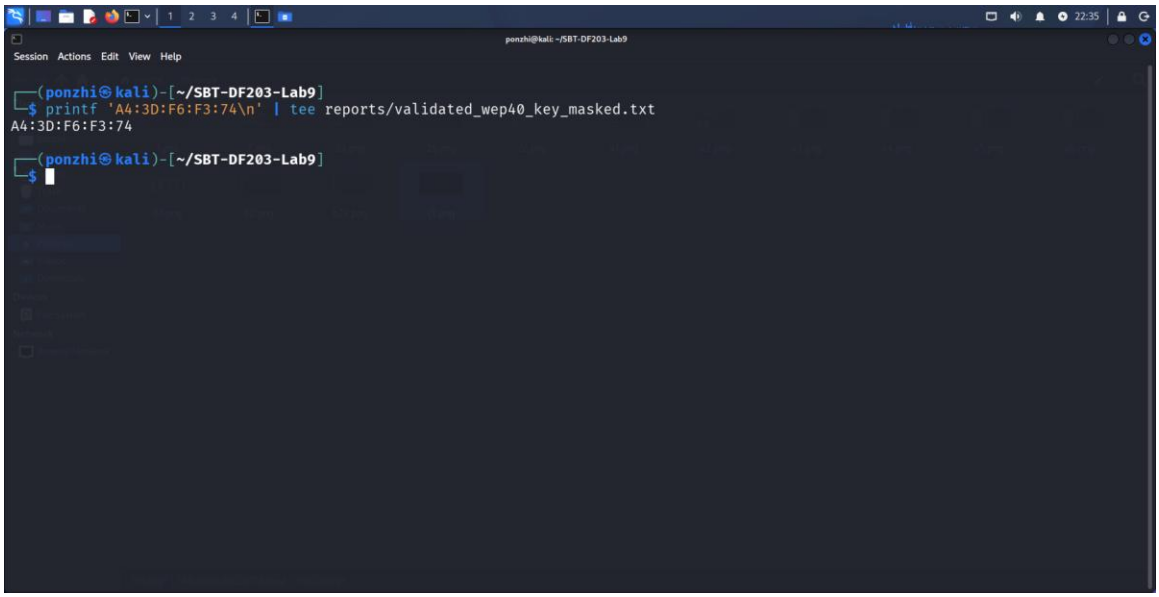
# Part C: Recover or Validate the WEP40 Key

Aircrack-ng was run directly against the capture to recover the WEP40 key using its built-in statistical (PTW) attack against the collected IVs.

```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
```




```
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```



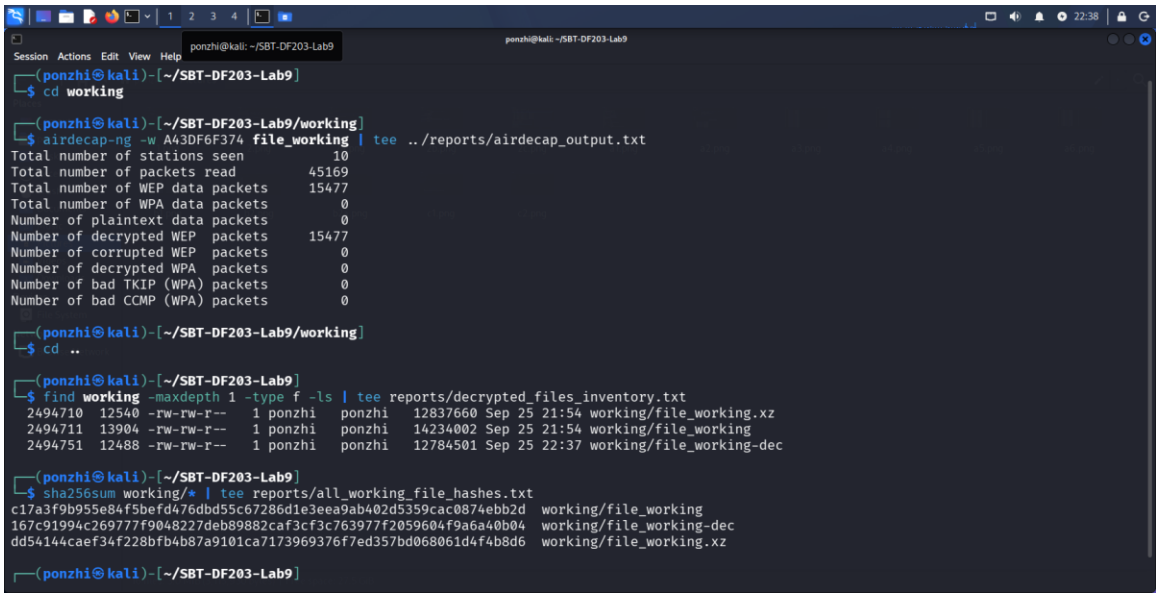

### Key-recovery summary

| | |
|---|---|
| **Target network (BSSID)** | 00:26:66:55:97:D6 |
| **ESSID** | cgnetwork |
| **Encryption** | WEP (15,477 IVs captured) |
| **Key length** | 40-bit (WEP40) |
| **Recovered key** | A4:3D:F6:F3:74 |
| **Validation** | Matches the slide-provided historical training key; Aircrack-ng reports “Decrypted correctly: 100%” |

## Part D: Decrypt the Capture Offline

With the key recovered and validated, we decrypted the capture offline using airdecap-ng. The key was supplied without colons, as required by the tool.

```
cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt
```




### Decryption results

| | |
|---|---|
| **Stations seen** | 10 |
| **Total packets read** | 45,169 |
| **WEP data packets** | 15,477 |
| **WPA data packets** | 0 |
| **Plaintext data packets** | 0 |
| **Decrypted WEP packets** | 15,477 (100%) |
| **Corrupted WEP packets** | 0 |
| **Decrypted capture filename** | working/file_working-dec |
| **Decrypted capture SHA-256** | 167c91994c269777f9048227deb89882caf3fc3c763977f2059604f9a6a40b04 |

# Part E: Extract Endpoints and Conversations

Endpoint and conversation analysis was performed against the decrypted capture produced in Section 11, now that IP/TCP/HTTP layers are visible.

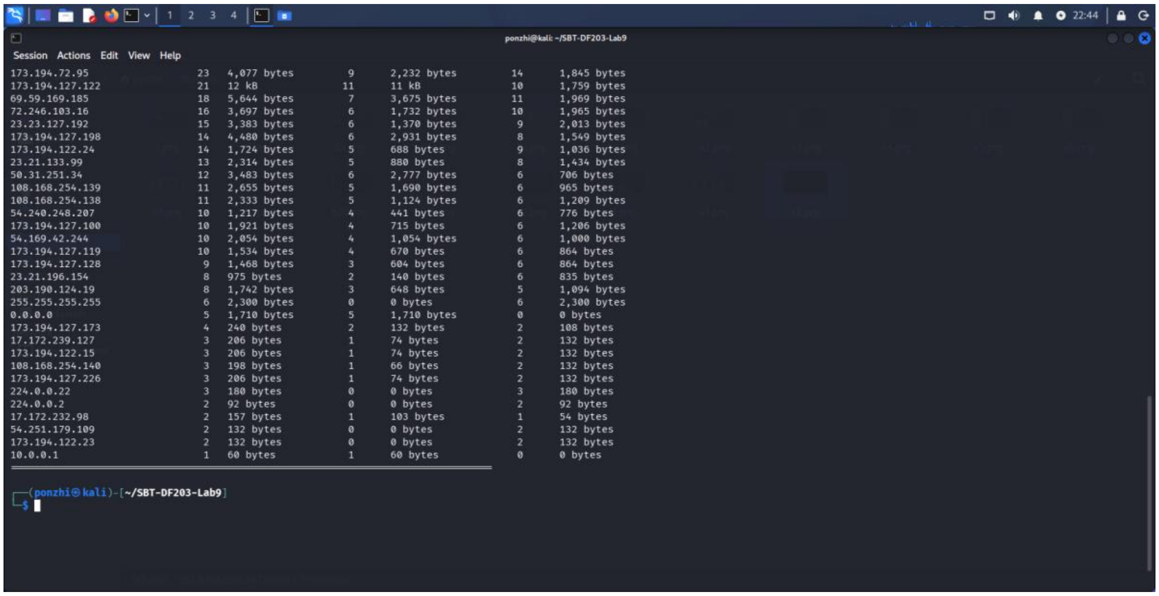
```
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
```

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

```
┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
Decrypted capture: working/file_working-dec

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
================================================================================
Ethernet Endpoints
Filter:<No Filter>
                      | Packets | |  Bytes  | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c            15407   12 MB          7076      616 kB          8331      11 MB
EFMNetworks_55:97:d4      15020   12 MB          8178      11 MB           6842      593 kB
ASUSTekCOMPU_2a:c2:7a       358   281 kB          223      270 kB           135      11 kB
IPv4mcast_fb                 42   6,648 bytes       0      0 bytes           42      6,648 bytes
IPv6mcast_fb                 36   7,022 bytes       0      0 bytes           36      7,022 bytes
Broadcast                    31   3,386 bytes       0      0 bytes           31      3,386 bytes
IPv4mcast_7f:ff:fa           31   11 kB             0      0 bytes           31      11 kB
IPv6mcast_16                 10   920 bytes         0      0 bytes           10      920 bytes
IPv6mcast_02                  9   630 bytes         0      0 bytes            9      630 bytes
IPv4mcast_16                  3   180 bytes         0      0 bytes            3      180 bytes
IPv4mcast_02                  2   92 bytes          0      0 bytes            2      92 bytes
NetisTechnol_48:e5:b4         1   42 bytes          0      0 bytes            1      42 bytes
NetisTechnol_48:a8:b5         1   42 bytes          0      0 bytes            1      42 bytes
08:10:77:92:7c:2f             1   42 bytes          0      0 bytes            1      42 bytes
TendaTechnol_58:c4:ba         1   42 bytes          0      0 bytes            1      42 bytes
IPv6mcast_ff:2a:c2:7a         1   78 bytes          0      0 bytes            1      78 bytes
================================================================================

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$
```

tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt

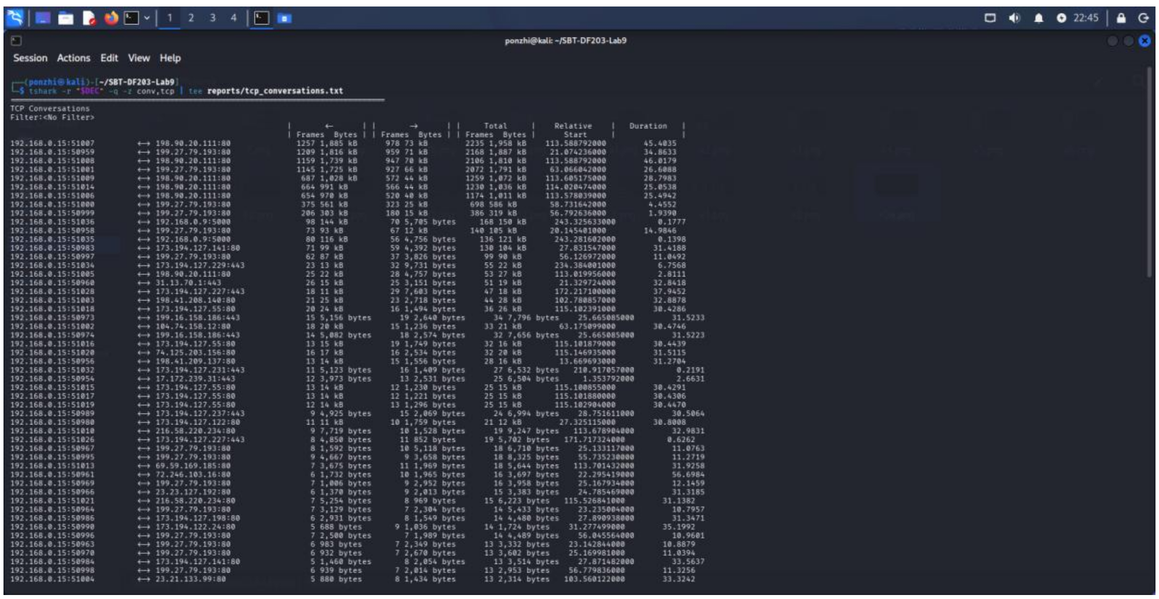
```
┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
================================================================================
IPv4 Endpoints
Filter:<No Filter>
                      | Packets | |  Bytes  | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
192.168.0.15              15309   12 MB          6995      608 kB          8314      11 MB
198.90.20.111              8057   6,916 kB       4446      6,637 kB        3611      279 kB
199.27.79.193              5694   4,825 kB       3131      4,603 kB        2563      221 kB
192.168.0.9                 329   277 kB          197      265 kB           132      11 kB
173.194.127.141             153   109 kB           80      102 kB            73      7,364 bytes
173.194.127.55              143   89 kB            71      82 kB             72      6,990 bytes
8.8.8.8                      81   10 kB            40      6,960 bytes       41      3,391 bytes
199.16.158.186               66   15 kB            29      10 kB             37      5,214 bytes
173.194.127.227              66   24 kB            26      15 kB             40      8,455 bytes
173.194.127.229              55   22 kB            23      13 kB             32      9,731 bytes
31.13.70.1                   51   19 kB            26      15 kB             25      3,151 bytes
173.241.248.220              44   6,848 bytes      20      3,122 bytes       24      3,726 bytes
198.41.208.140               44   28 kB            21      25 kB             23      2,718 bytes
224.0.0.251                  42   6,648 bytes       0      0 bytes           42      6,648 bytes
173.194.127.237              36   9,526 bytes      14      6,435 bytes       22      3,091 bytes
216.58.220.234               34   15 kB            16      12 kB             18      2,497 bytes
192.168.0.1                  33   12 kB            33      12 kB              0      0 bytes
173.241.248.219              33   5,733 bytes      14      2,807 bytes       19      2,926 bytes
104.74.158.12                33   21 kB            18      20 kB             15      1,236 bytes
74.125.203.156               32   20 kB            16      17 kB             16      2,534 bytes
239.255.255.250              31   11 kB             0      0 bytes           31      11 kB
173.194.127.231              31   6,816 bytes      12      5,197 bytes       19      1,619 bytes
198.41.209.137               28   16 kB            13      14 kB             15      1,556 bytes
17.172.239.31                25   6,504 bytes      12      3,973 bytes       13      2,531 bytes
173.194.72.95                23   4,077 bytes       9      2,232 bytes       14      1,845 bytes
```



| | | | | | | |
|---|---|---|---|---|---|---|
| 173.194.72.95 | 23 | 4,077 bytes | 9 | 2,232 bytes | 14 | 1,845 bytes |
| 173.194.127.122 | 21 | 12 kB | 11 | 11 kB | 10 | 1,759 bytes |
| 69.59.169.185 | 18 | 5,644 bytes | 7 | 3,675 bytes | 11 | 1,969 bytes |
| 72.246.103.16 | 16 | 3,697 bytes | 6 | 1,732 bytes | 10 | 1,965 bytes |
| 23.23.127.192 | 15 | 3,383 bytes | 6 | 1,370 bytes | 9 | 2,013 bytes |
| 173.194.127.198 | 14 | 4,480 bytes | 6 | 2,931 bytes | 8 | 1,549 bytes |
| 173.194.122.24 | 14 | 1,724 bytes | 5 | 688 bytes | 9 | 1,036 bytes |
| 23.21.133.99 | 13 | 2,314 bytes | 5 | 880 bytes | 8 | 1,434 bytes |
| 50.31.251.34 | 12 | 3,483 bytes | 6 | 2,777 bytes | 6 | 706 bytes |
| 108.168.254.139 | 11 | 2,655 bytes | 5 | 1,690 bytes | 6 | 965 bytes |
| 108.168.254.138 | 11 | 2,333 bytes | 5 | 1,124 bytes | 6 | 1,209 bytes |
| 54.240.248.207 | 10 | 1,217 bytes | 4 | 441 bytes | 6 | 776 bytes |
| 173.194.127.100 | 10 | 1,921 bytes | 4 | 715 bytes | 6 | 1,206 bytes |
| 54.169.42.244 | 10 | 2,054 bytes | 4 | 1,054 bytes | 6 | 1,000 bytes |
| 173.194.127.119 | 10 | 1,534 bytes | 4 | 670 bytes | 6 | 864 bytes |
| 173.194.127.128 | 9 | 1,468 bytes | 3 | 604 bytes | 6 | 864 bytes |
| 23.21.196.154 | 8 | 975 bytes | 2 | 140 bytes | 6 | 835 bytes |
| 203.190.124.19 | 8 | 1,742 bytes | 3 | 648 bytes | 5 | 1,094 bytes |
| 255.255.255.255 | 6 | 2,300 bytes | 0 | 0 bytes | 6 | 2,300 bytes |
| 0.0.0.0 | 5 | 1,710 bytes | 5 | 1,710 bytes | 0 | 0 bytes |
| 173.194.127.173 | 4 | 240 bytes | 2 | 132 bytes | 2 | 108 bytes |
| 17.172.239.127 | 3 | 206 bytes | 1 | 74 bytes | 2 | 132 bytes |
| 173.194.122.15 | 3 | 206 bytes | 1 | 74 bytes | 2 | 132 bytes |
| 108.168.254.140 | 3 | 198 bytes | 1 | 66 bytes | 2 | 132 bytes |
| 173.194.127.226 | 3 | 206 bytes | 1 | 74 bytes | 2 | 132 bytes |
| 224.0.0.22 | 3 | 180 bytes | 0 | 0 bytes | 3 | 180 bytes |
| 224.0.0.2 | 2 | 92 bytes | 0 | 0 bytes | 2 | 92 bytes |
| 17.172.232.98 | 2 | 157 bytes | 1 | 103 bytes | 1 | 54 bytes |
| 54.251.179.109 | 2 | 132 bytes | 0 | 0 bytes | 2 | 132 bytes |
| 173.194.122.23 | 2 | 132 bytes | 0 | 0 bytes | 2 | 132 bytes |
| 10.0.0.1 | 1 | 60 bytes | 1 | 60 bytes | 0 | 0 bytes |

(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
$

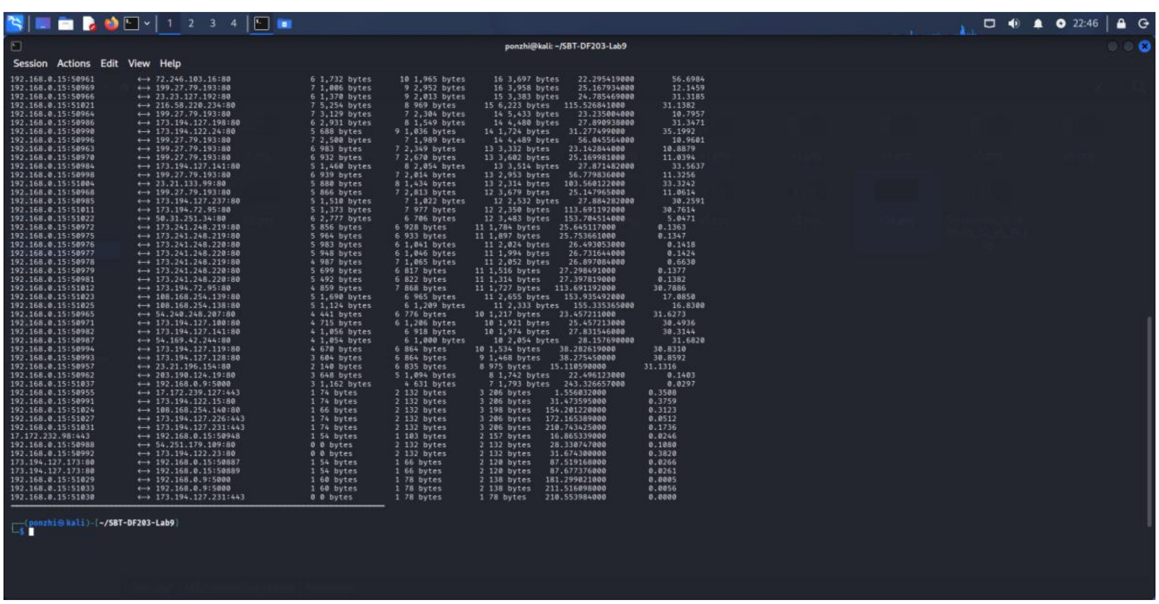
```
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
```



(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

TCP Conversations
Filter:<No Filter>

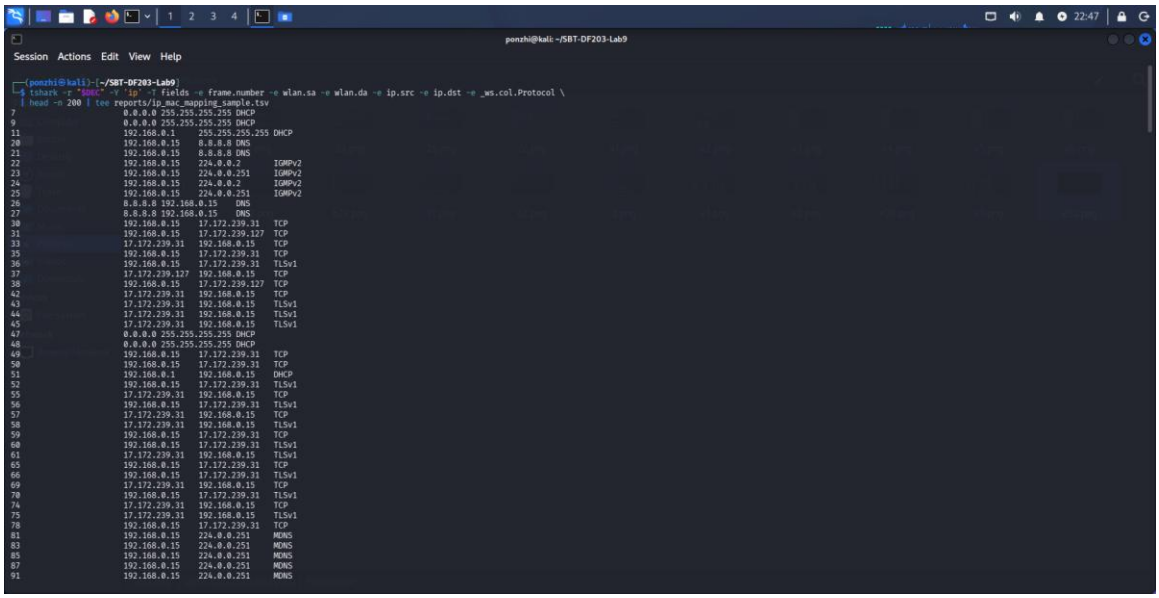
| | | ← Frames | ← Bytes | → Frames | → Bytes | Total Frames | Total Bytes | Relative Start | Duration |
|---|---|---|---|---|---|---|---|---|---|
| 192.168.0.15:51007 | ↔ 198.90.20.111:80 | 1257 | 1,885 kB | 978 | 73 kB | 2235 | 1,958 kB | 113.588792000 | 45.4035 |
| 192.168.0.15:50959 | ↔ 199.27.79.193:80 | 1209 | 1,816 kB | 959 | 71 kB | 2168 | 1,887 kB | 21.074236000 | 34.8633 |
| 192.168.0.15:51008 | ↔ 198.90.20.111:80 | 1159 | 1,739 kB | 947 | 70 kB | 2106 | 1,810 kB | 113.588792000 | 46.0179 |
| 192.168.0.15:51001 | ↔ 199.27.79.193:80 | 1145 | 1,725 kB | 927 | 66 kB | 2072 | 1,791 kB | 63.066042000 | 26.6088 |
| 192.168.0.15:51009 | ↔ 198.90.20.111:80 | 687 | 1,028 kB | 572 | 44 kB | 1259 | 1,072 kB | 113.605175000 | 28.7983 |
| 192.168.0.15:51014 | ↔ 198.90.20.111:80 | 664 | 991 kB | 566 | 44 kB | 1230 | 1,036 kB | 114.020474000 | 25.0538 |
| 192.168.0.15:51006 | ↔ 198.90.20.111:80 | 654 | 970 kB | 520 | 40 kB | 1174 | 1,011 kB | 113.578039000 | 25.4942 |
| 192.168.0.15:51000 | ↔ 199.27.79.193:80 | 375 | 561 kB | 323 | 25 kB | 698 | 586 kB | 58.731642000 | 4.4552 |
| 192.168.0.15:50999 | ↔ 199.27.79.193:80 | 206 | 303 kB | 180 | 15 kB | 386 | 319 kB | 56.792636000 | 1.9390 |
| 192.168.0.15:51036 | ↔ 192.168.0.9:5000 | 98 | 144 kB | 70 | 5,705 bytes | 168 | 150 kB | 243.325633000 | 0.1777 |
| 192.168.0.15:50958 | ↔ 199.27.79.193:80 | 73 | 93 kB | 67 | 12 kB | 140 | 105 kB | 20.145401000 | 14.9846 |
| 192.168.0.15:51035 | ↔ 192.168.0.9:5000 | 80 | 116 kB | 56 | 4,756 bytes | 136 | 121 kB | 243.281602000 | 0.1398 |
| 192.168.0.15:50983 | ↔ 173.194.127.141:80 | 71 | 99 kB | 59 | 4,392 bytes | 130 | 104 kB | 27.831547000 | 31.4188 |
| 192.168.0.15:50997 | ↔ 199.27.79.193:80 | 62 | 87 kB | 37 | 3,826 bytes | 99 | 90 kB | 56.126972000 | 11.0492 |
| 192.168.0.15:51034 | ↔ 173.194.127.229:443 | 23 | 13 kB | 32 | 9,731 bytes | 55 | 22 kB | 234.384001000 | 6.7568 |
| 192.168.0.15:51005 | ↔ 198.90.20.111:80 | 25 | 22 kB | 28 | 4,757 bytes | 53 | 27 kB | 113.019956000 | 2.8111 |
| 192.168.0.15:50960 | ↔ 31.13.70.1:443 | 26 | 15 kB | 25 | 3,151 bytes | 51 | 19 kB | 21.329724000 | 32.8418 |
| 192.168.0.15:51028 | ↔ 173.194.127.227:443 | 18 | 11 kB | 29 | 7,603 bytes | 47 | 18 kB | 172.217100000 | 37.9452 |
| 192.168.0.15:51003 | ↔ 198.41.208.140:80 | 21 | 25 kB | 23 | 2,718 bytes | 44 | 28 kB | 102.780857000 | 32.8878 |
| 192.168.0.15:51018 | ↔ 173.194.127.55:80 | 20 | 24 kB | 16 | 1,494 bytes | 36 | 26 kB | 115.102391000 | 30.4286 |
| 192.168.0.15:50973 | ↔ 199.16.158.186:443 | 15 | 5,156 bytes | 19 | 2,640 bytes | 34 | 7,796 bytes | 25.665085000 | 31.5233 |
| 192.168.0.15:51002 | ↔ 104.74.158.12:80 | 18 | 20 kB | 15 | 1,236 bytes | 33 | 21 kB | 63.175099000 | 30.4746 |
| 192.168.0.15:50974 | ↔ 199.16.158.186:443 | 14 | 5,082 bytes | 18 | 2,574 bytes | 32 | 7,656 bytes | 25.665085000 | 31.5223 |
| 192.168.0.15:51016 | ↔ 173.194.127.55:80 | 13 | 15 kB | 19 | 1,749 bytes | 32 | 16 kB | 115.101879000 | 30.4439 |
| 192.168.0.15:51020 | ↔ 74.125.203.156:80 | 16 | 17 kB | 16 | 2,534 bytes | 32 | 20 kB | 115.146935000 | 31.5115 |
| 192.168.0.15:50956 | ↔ 198.41.209.137:80 | 13 | 14 kB | 15 | 1,556 bytes | 28 | 16 kB | 13.669693000 | 31.2704 |
| 192.168.0.15:51032 | ↔ 173.194.127.231:443 | 11 | 5,123 bytes | 16 | 1,409 bytes | 27 | 6,532 bytes | 210.917057000 | 0.2191 |
| 192.168.0.15:50954 | ↔ 17.172.239.31:443 | 12 | 3,973 bytes | 13 | 2,531 bytes | 25 | 6,504 bytes | 1.353792000 | 2.6631 |
| 192.168.0.15:51015 | ↔ 173.194.127.55:80 | 13 | 14 kB | 12 | 1,230 bytes | 25 | 15 kB | 115.100855000 | 30.4291 |
| 192.168.0.15:51017 | ↔ 173.194.127.55:80 | 13 | 14 kB | 12 | 1,221 bytes | 25 | 15 kB | 115.101880000 | 30.4306 |
| 192.168.0.15:51019 | ↔ 173.194.127.55:80 | 12 | 14 kB | 13 | 1,296 bytes | 25 | 15 kB | 115.102904000 | 30.4470 |
| 192.168.0.15:50989 | ↔ 173.194.127.237:443 | 9 | 4,925 bytes | 15 | 2,069 bytes | 24 | 6,994 bytes | 28.751611000 | 30.5064 |
| 192.168.0.15:50980 | ↔ 173.194.127.122:80 | 11 | 11 kB | 10 | 1,759 bytes | 21 | 12 kB | 27.325115000 | 30.8008 |
| 192.168.0.15:51010 | ↔ 216.58.220.234:80 | 9 | 7,719 bytes | 10 | 1,528 bytes | 19 | 9,247 bytes | 113.678904000 | 32.9831 |
| 192.168.0.15:51026 | ↔ 173.194.127.227:443 | 8 | 4,850 bytes | 11 | 852 bytes | 19 | 5,702 bytes | 171.717324000 | 0.6262 |
| 192.168.0.15:50967 | ↔ 199.27.79.193:80 | 8 | 1,592 bytes | 10 | 5,118 bytes | 18 | 6,710 bytes | 25.133117000 | 11.0763 |
| 192.168.0.15:50995 | ↔ 199.27.79.193:80 | 9 | 4,667 bytes | 9 | 3,658 bytes | 18 | 8,325 bytes | 55.735230000 | 11.2719 |
| 192.168.0.15:51013 | ↔ 69.59.169.185:80 | 7 | 3,675 bytes | 11 | 1,969 bytes | 18 | 5,644 bytes | 113.701432000 | 31.9258 |
| 192.168.0.15:50961 | ↔ 72.246.103.16:80 | 6 | 1,732 bytes | 10 | 1,965 bytes | 16 | 3,697 bytes | 22.295419000 | 56.6984 |
| 192.168.0.15:50969 | ↔ 199.27.79.193:80 | 7 | 1,006 bytes | 9 | 2,952 bytes | 16 | 3,958 bytes | 25.167934000 | 12.1459 |
| 192.168.0.15:50966 | ↔ 23.23.127.192:80 | 6 | 1,370 bytes | 9 | 2,013 bytes | 15 | 3,383 bytes | 24.785469000 | 31.3185 |
| 192.168.0.15:51021 | ↔ 216.58.220.234:80 | 7 | 5,254 bytes | 8 | 969 bytes | 15 | 6,223 bytes | 115.526841000 | 31.1382 |
| 192.168.0.15:50964 | ↔ 199.27.79.193:80 | 7 | 3,129 bytes | 7 | 2,304 bytes | 14 | 5,433 bytes | 23.235004000 | 10.7957 |
| 192.168.0.15:50986 | ↔ 173.194.127.198:80 | 6 | 2,931 bytes | 8 | 1,549 bytes | 14 | 4,480 bytes | 27.890938000 | 31.3471 |
| 192.168.0.15:50990 | ↔ 173.194.122.24:80 | 5 | 688 bytes | 9 | 1,036 bytes | 14 | 1,724 bytes | 31.277499000 | 35.1992 |
| 192.168.0.15:50996 | ↔ 199.27.79.193:80 | 7 | 2,500 bytes | 7 | 1,989 bytes | 14 | 4,489 bytes | 56.045564000 | 10.9601 |
| 192.168.0.15:50963 | ↔ 199.27.79.193:80 | 6 | 983 bytes | 7 | 2,349 bytes | 13 | 3,332 bytes | 23.142844000 | 10.8879 |
| 192.168.0.15:50970 | ↔ 199.27.79.193:80 | 6 | 932 bytes | 7 | 2,670 bytes | 13 | 3,602 bytes | 25.169981000 | 11.0394 |
| 192.168.0.15:50984 | ↔ 173.194.127.141:80 | 5 | 1,460 bytes | 8 | 2,054 bytes | 13 | 3,514 bytes | 27.871482000 | 33.5637 |
| 192.168.0.15:50998 | ↔ 199.27.79.193:80 | 6 | 939 bytes | 7 | 2,014 bytes | 13 | 2,953 bytes | 56.779836000 | 11.3256 |
| 192.168.0.15:51004 | ↔ 23.21.133.99:80 | 5 | 880 bytes | 8 | 1,434 bytes | 13 | 2,314 bytes | 103.560122000 | 33.3242 |

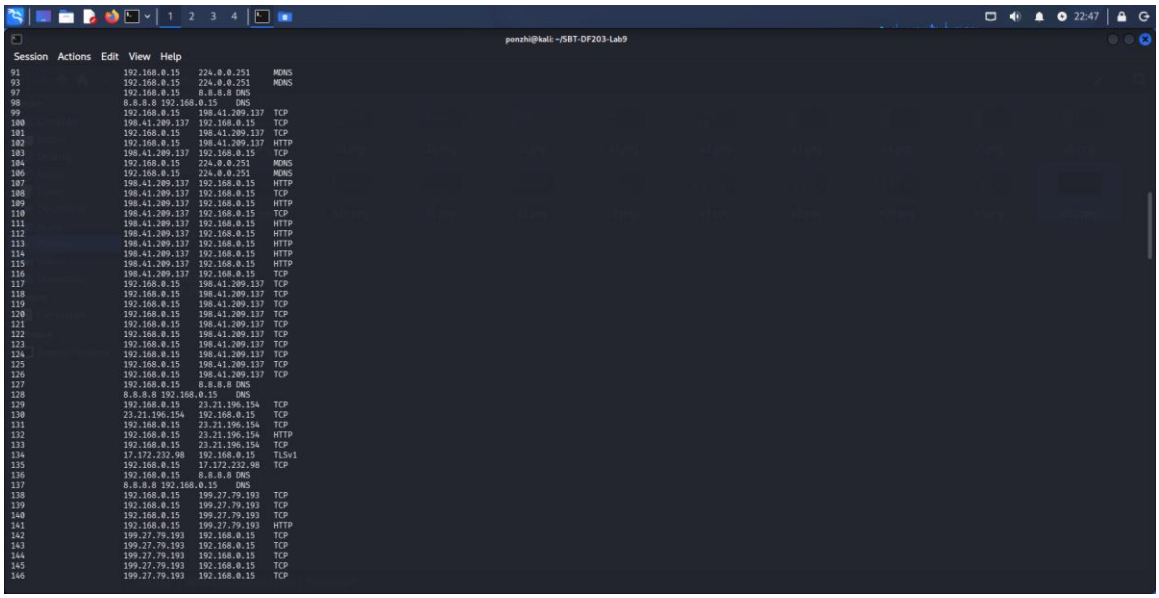


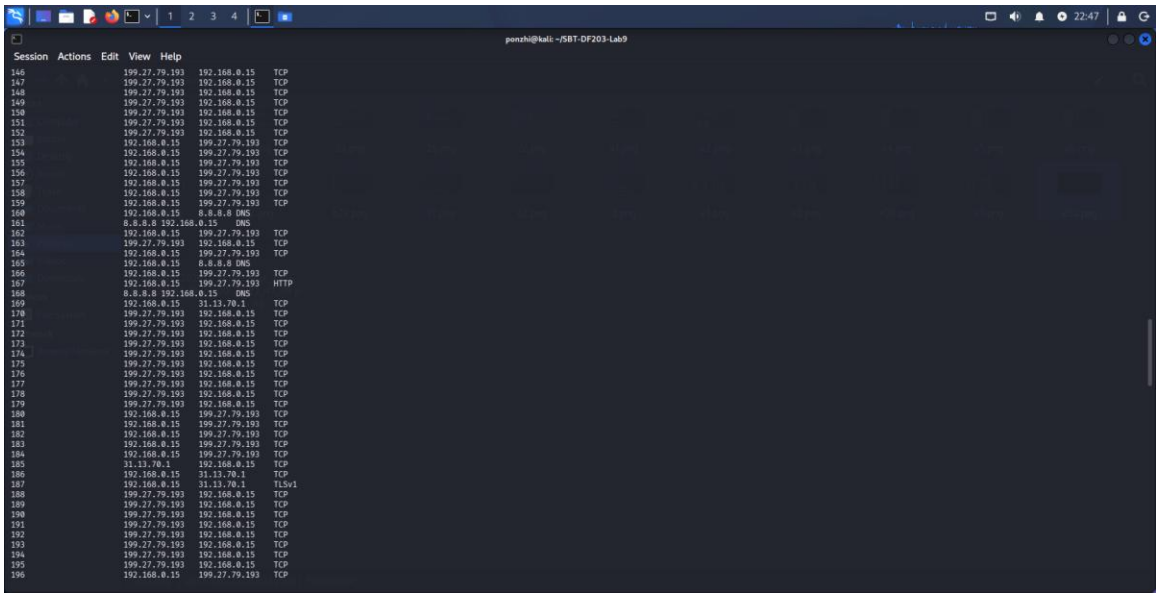
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 192.168.0.15:50961 | ↔ 72.246.103.16:80 | 6 | 1,732 bytes | 10 | 1,965 bytes | 16 | 3,697 bytes | 22.295419000 | 56.6984 |
| 192.168.0.15:50969 | ↔ 199.27.79.193:80 | 7 | 1,006 bytes | 9 | 2,952 bytes | 16 | 3,958 bytes | 25.167934000 | 12.1459 |
| 192.168.0.15:50966 | ↔ 23.23.127.192:80 | 6 | 1,370 bytes | 9 | 2,013 bytes | 15 | 3,383 bytes | 24.785469000 | 31.3185 |
| 192.168.0.15:51021 | ↔ 216.58.220.234:80 | 7 | 5,254 bytes | 8 | 969 bytes | 15 | 6,223 bytes | 115.526841000 | 31.1382 |
| 192.168.0.15:50964 | ↔ 199.27.79.193:80 | 7 | 3,129 bytes | 7 | 2,304 bytes | 14 | 5,433 bytes | 23.235004000 | 10.7957 |
| 192.168.0.15:50986 | ↔ 173.194.127.198:80 | 6 | 2,931 bytes | 8 | 1,549 bytes | 14 | 4,480 bytes | 27.890938000 | 31.3471 |
| 192.168.0.15:50990 | ↔ 173.194.122.24:80 | 5 | 688 bytes | 9 | 1,036 bytes | 14 | 1,724 bytes | 31.277499000 | 35.1992 |
| 192.168.0.15:50996 | ↔ 199.27.79.193:80 | 7 | 2,500 bytes | 7 | 1,989 bytes | 14 | 4,489 bytes | 56.045564000 | 10.9601 |
| 192.168.0.15:50963 | ↔ 199.27.79.193:80 | 6 | 983 bytes | 7 | 2,349 bytes | 13 | 3,332 bytes | 23.142844000 | 10.8879 |
| 192.168.0.15:50970 | ↔ 199.27.79.193:80 | 6 | 932 bytes | 7 | 2,670 bytes | 13 | 3,602 bytes | 25.169981000 | 11.0394 |
| 192.168.0.15:50984 | ↔ 173.194.127.141:80 | 5 | 1,460 bytes | 8 | 2,054 bytes | 13 | 3,514 bytes | 27.871482000 | 33.5637 |
| 192.168.0.15:50998 | ↔ 199.27.79.193:80 | 6 | 939 bytes | 7 | 2,014 bytes | 13 | 2,953 bytes | 56.779836000 | 11.3256 |
| 192.168.0.15:51004 | ↔ 23.21.133.99:80 | 5 | 880 bytes | 8 | 1,434 bytes | 13 | 2,314 bytes | 103.560122000 | 33.3242 |
| 192.168.0.15:50968 | ↔ 199.27.79.193:80 | 5 | 866 bytes | 7 | 2,813 bytes | 12 | 3,679 bytes | 25.147965000 | 11.0614 |
| 192.168.0.15:50985 | ↔ 173.194.127.237:80 | 5 | 1,510 bytes | 7 | 1,022 bytes | 12 | 2,532 bytes | 27.884282000 | 30.2591 |
| 192.168.0.15:51011 | ↔ 173.194.72.95:80 | 5 | 1,373 bytes | 7 | 977 bytes | 12 | 2,350 bytes | 113.691192000 | 30.7614 |
| 192.168.0.15:51022 | ↔ 50.31.251.34:80 | 6 | 2,777 bytes | 6 | 706 bytes | 12 | 3,483 bytes | 153.704514000 | 5.0471 |
| 192.168.0.15:50972 | ↔ 173.241.248.219:80 | 5 | 856 bytes | 6 | 928 bytes | 11 | 1,784 bytes | 25.645117000 | 0.1363 |
| 192.168.0.15:50975 | ↔ 173.241.248.219:80 | 5 | 964 bytes | 6 | 933 bytes | 11 | 1,897 bytes | 25.753661000 | 0.1347 |
| 192.168.0.15:50976 | ↔ 173.241.248.220:80 | 5 | 983 bytes | 6 | 1,041 bytes | 11 | 2,024 bytes | 26.493053000 | 0.1418 |
| 192.168.0.15:50977 | ↔ 173.241.248.220:80 | 5 | 948 bytes | 6 | 1,046 bytes | 11 | 1,994 bytes | 26.731644000 | 0.1424 |
| 192.168.0.15:50978 | ↔ 173.241.248.219:80 | 4 | 987 bytes | 7 | 1,065 bytes | 11 | 2,052 bytes | 26.897084000 | 0.6630 |
| 192.168.0.15:50979 | ↔ 173.241.248.220:80 | 5 | 699 bytes | 6 | 817 bytes | 11 | 1,516 bytes | 27.298491000 | 0.1377 |
| 192.168.0.15:50981 | ↔ 173.241.248.220:80 | 5 | 492 bytes | 6 | 822 bytes | 11 | 1,314 bytes | 27.397819000 | 0.1382 |
| 192.168.0.15:51012 | ↔ 173.194.72.95:80 | 4 | 859 bytes | 7 | 868 bytes | 11 | 1,727 bytes | 113.691192000 | 30.7886 |
| 192.168.0.15:51023 | ↔ 108.168.254.139:80 | 5 | 1,690 bytes | 6 | 965 bytes | 11 | 2,655 bytes | 153.935492000 | 17.0850 |
| 192.168.0.15:51025 | ↔ 108.168.254.138:80 | 5 | 1,124 bytes | 6 | 1,209 bytes | 11 | 2,333 bytes | 155.335365000 | 16.8300 |
| 192.168.0.15:50965 | ↔ 54.240.248.207:80 | 4 | 441 bytes | 6 | 776 bytes | 10 | 1,217 bytes | 23.457211000 | 31.6273 |
| 192.168.0.15:50971 | ↔ 173.194.127.100:80 | 4 | 715 bytes | 6 | 1,206 bytes | 10 | 1,921 bytes | 25.457213000 | 30.4936 |
| 192.168.0.15:50982 | ↔ 173.194.127.141:80 | 4 | 1,056 bytes | 6 | 918 bytes | 10 | 1,974 bytes | 27.831546000 | 30.3144 |
| 192.168.0.15:50987 | ↔ 54.169.42.244:80 | 4 | 1,054 bytes | 6 | 1,000 bytes | 10 | 2,054 bytes | 28.157690000 | 31.6820 |
| 192.168.0.15:50994 | ↔ 173.194.127.119:80 | 4 | 670 bytes | 6 | 864 bytes | 10 | 1,534 bytes | 38.282619000 | 30.8310 |
| 192.168.0.15:50993 | ↔ 173.194.127.128:80 | 3 | 604 bytes | 6 | 864 bytes | 9 | 1,468 bytes | 38.275450000 | 30.8592 |
| 192.168.0.15:50957 | ↔ 23.21.196.154:80 | 2 | 140 bytes | 6 | 835 bytes | 8 | 975 bytes | 15.110590000 | 31.1316 |
| 192.168.0.15:50962 | ↔ 203.190.124.19:80 | 3 | 648 bytes | 5 | 1,094 bytes | 8 | 1,742 bytes | 22.496123000 | 0.1403 |
| 192.168.0.15:51037 | ↔ 192.168.0.9:5000 | 3 | 1,162 bytes | 4 | 631 bytes | 7 | 1,793 bytes | 243.326657000 | 0.0297 |
| 192.168.0.15:50955 | ↔ 17.172.239.127:443 | 1 | 74 bytes | 2 | 132 bytes | 3 | 206 bytes | 1.556032000 | 0.3508 |
| 192.168.0.15:50991 | ↔ 173.194.122.15:80 | 1 | 74 bytes | 2 | 132 bytes | 3 | 206 bytes | 31.473595000 | 0.3759 |
| 192.168.0.15:51024 | ↔ 108.168.254.140:80 | 1 | 66 bytes | 2 | 132 bytes | 3 | 198 bytes | 154.201220000 | 0.3123 |
| 192.168.0.15:51027 | ↔ 173.194.127.226:443 | 1 | 74 bytes | 2 | 132 bytes | 3 | 206 bytes | 172.165389000 | 0.0512 |
| 192.168.0.15:51031 | ↔ 173.194.127.231:443 | 1 | 74 bytes | 2 | 132 bytes | 3 | 206 bytes | 210.743425000 | 0.1736 |
| 17.172.232.98:443 | ↔ 192.168.0.15:50948 | 1 | 54 bytes | 1 | 103 bytes | 2 | 157 bytes | 16.865339000 | 0.0246 |
| 192.168.0.15:50988 | ↔ 54.251.179.109:80 | 0 | 0 bytes | 2 | 132 bytes | 2 | 132 bytes | 28.330747000 | 0.1080 |
| 192.168.0.15:50992 | ↔ 173.194.122.23:80 | 0 | 0 bytes | 2 | 132 bytes | 2 | 132 bytes | 31.674300000 | 0.3820 |
| 173.194.127.173:80 | ↔ 192.168.0.15:50887 | 1 | 54 bytes | 1 | 66 bytes | 2 | 120 bytes | 87.519168000 | 0.0266 |
| 173.194.127.173:80 | ↔ 192.168.0.15:50889 | 1 | 54 bytes | 1 | 66 bytes | 2 | 120 bytes | 87.677376000 | 0.0261 |
| 192.168.0.15:51029 | ↔ 192.168.0.9:5000 | 1 | 60 bytes | 1 | 78 bytes | 2 | 138 bytes | 181.299021000 | 0.0005 |
| 192.168.0.15:51033 | ↔ 192.168.0.9:5000 | 1 | 60 bytes | 1 | 78 bytes | 2 | 138 bytes | 211.516098000 | 0.0056 |
| 192.168.0.15:51030 | ↔ 173.194.127.231:443 | 0 | 0 bytes | 1 | 78 bytes | 1 | 78 bytes | 210.553984000 | 0.0000 |

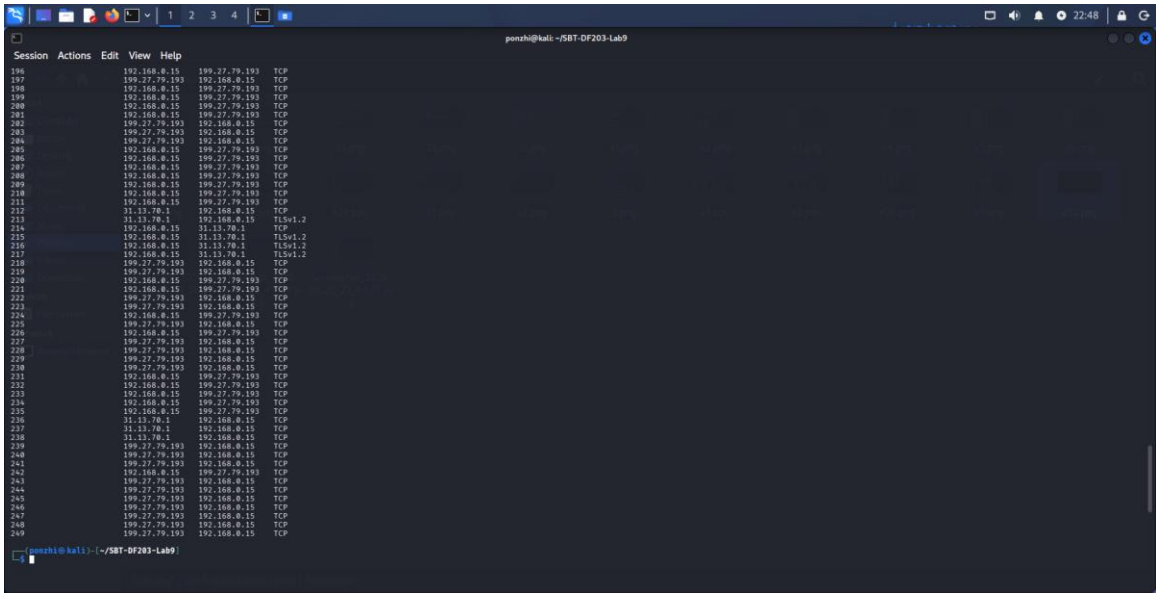
(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
$

```
tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
  | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```






## Ethernet endpoint summary

| Endpoint (vendor:suffix) | Packets | Bytes | Role |
|---|---|---|---|
| Apple_68:96:7c (f0:f6:1c:68:96:7c) | 15,407 | 12 MB | Host/client station (matches WEP source MAC observed earlier) |
| EFMNetworks_55:97:d4 (00:26:66:55:97:d4) | 15,020 | 11 MB | Wireless access point/gateway (same OUI as BSSID 00:26:66:55:97:d6) |
| ASUSTekCOMPU_2a:c2:7a | 358 | 281 kB | Second local station on the network |
| Broadcast / multicast (IPv4mcast, IPv6mcast, Broadcast) | ≤ 42 each | ≤ 11 kB each | DHCP, mDNS, and multicast discovery traffic |

## Top IPv4 endpoints

| IP Address | Packets | Bytes | Notes |
|---|---|---|---|
| 192.168.0.15 | 15,309 | 12 MB | Host — local station, generates almost all outbound traffic |
| 198.90.20.111 | 8,057 | 6,916 kB | Heaviest remote endpoint — primary HTTP session |
| 199.27.79.193 | 5,694 | 4,825 kB | Second-heaviest remote endpoint — sustained HTTP session |
| 192.168.0.9 | 329 | 277 kB | Secondary local host on the same subnet |
| 8.8.8.8 | 81 | 10 kB | Google Public DNS resolver |
| 173.194.x.x range | various | various | Google-hosted services (ads/analytics) |
| 17.172.x.x range | various | various | Apple push-notification / CDN infrastructure |
| 31.13.70.1 | 51 | 19 kB | Facebook (TLSv1.2 session, likely a share widget) |

## Top TCP conversations

| Conversation | Frames (↔) | Bytes (↔) | Duration (s) |
|---|---|---|---|
| 192.168.0.15:51007 ↔ 198.90.20.111:80 | 2,235 | 1,958 kB | 45.40 |
| 192.168.0.15:50959 ↔ 199.27.79.193:80 | 2,168 | 1,887 kB | 34.86 |
| 192.168.0.15:51008 ↔ 198.90.20.111:80 | 2,106 | 1,810 kB | 46.02 |
| 192.168.0.15:51001 ↔ 198.90.20.111:80 | 2,072 | 1,791 kB | 26.61 |
| 192.168.0.15:51009 ↔ 198.90.20.111:80 | 1,259 | 1,072 kB | 28.80 |

## Host/gateway identification

| | |
|---|---|
| **Host IP** | 192.168.0.15 |
| **Host MAC** | f0:f6:1c:68:96:7c (Apple_68:96:7c) — confirmed via ARP announcement "Apple_68:96:7c → Broadcast: ARP Announcement for 192.168.0.15" and matches the station MAC seen in the WEP-protected frames in Section 9 |
| **Gateway/AP IP** | 192.168.0.1 |
| **Gateway/AP MAC** | 00:26:66:55:97:d4 (EFMNetworks_55:97:d4) — same OUI/device as BSSID 00:26:66:55:97:d6 identified by Aircrack-ng |
| **Server IP(s) observed** | No remote server's own MAC is visible in this capture — all off-subnet traffic is forwarded through the gateway (00:26:66:55:97:d4), so only IP addresses are recoverable for remote hosts. The two heaviest remote IPs, 198.90.20.111 and 199.27.79.193, account for the bulk of the HTTP traffic; 198.41.209.137 and 23.21.196.154 were also observed serving HTTP content. |

## Key protocols observed

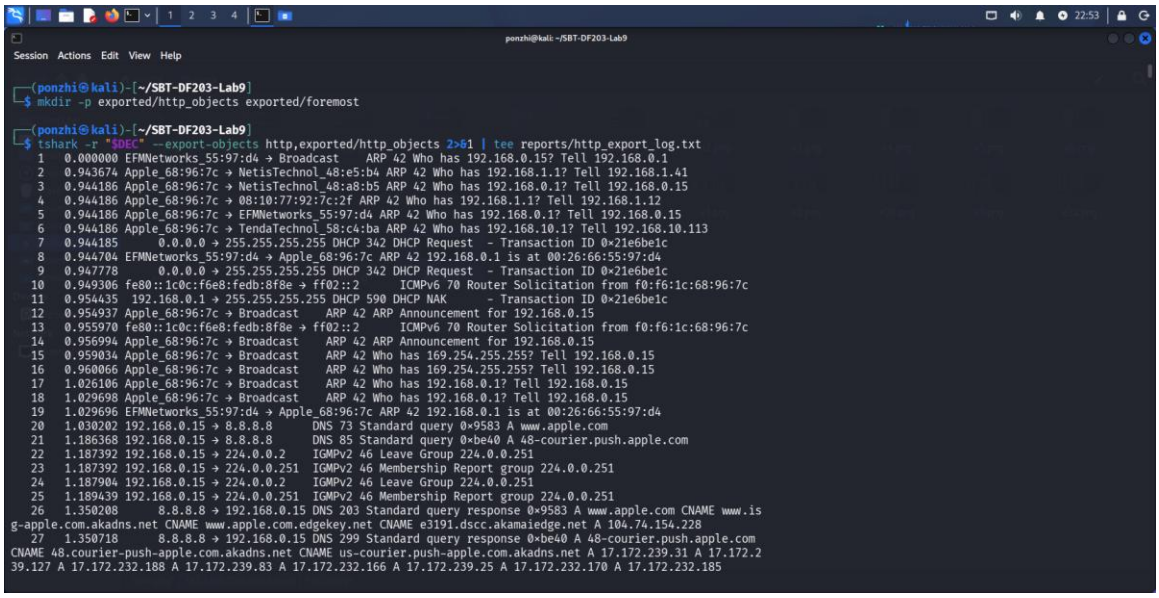
| Protocol | Purpose in this capture |
|---|---|
| ARP | Address resolution between host (192.168.0.15) and gateway (192.168.0.1) |

| Protocol | Purpose in this capture |
|---|---|
| DHCP | Host lease acquisition (Transaction ID 0x21e6be1c/0x21e6be1d), including a DHCP NAK/renegotiation |
| DNS | Lookups to 8.8.8.8 (e.g. www.apple.com, 48-courier.push.apple.com) |
| ICMPv6 | Router solicitation / multicast listener reports from the host's IPv6 stack |
| IGMPv2 / MDNS | Local multicast group membership and mDNS discovery (224.0.0.251 etc.) |
| TCP | Transport for the HTTP and TLS sessions; visible 3-way handshakes and window scaling |
| TLSv1 / TLSv1.2 | Encrypted sessions to Apple push infrastructure (17.172.x.x) and Facebook (31.13.70.1) |
| HTTP | Plaintext web session(s) to 198.41.209.137, 199.27.79.193, 23.21.196.154 — source of the objects extracted in Section 13 |

# Part F: Extract Images, HTML and Other Objects

Wireshark/tshark's HTTP object export was run first, followed by foremost as a generic file-carving pass, per the lab's two-method requirement.

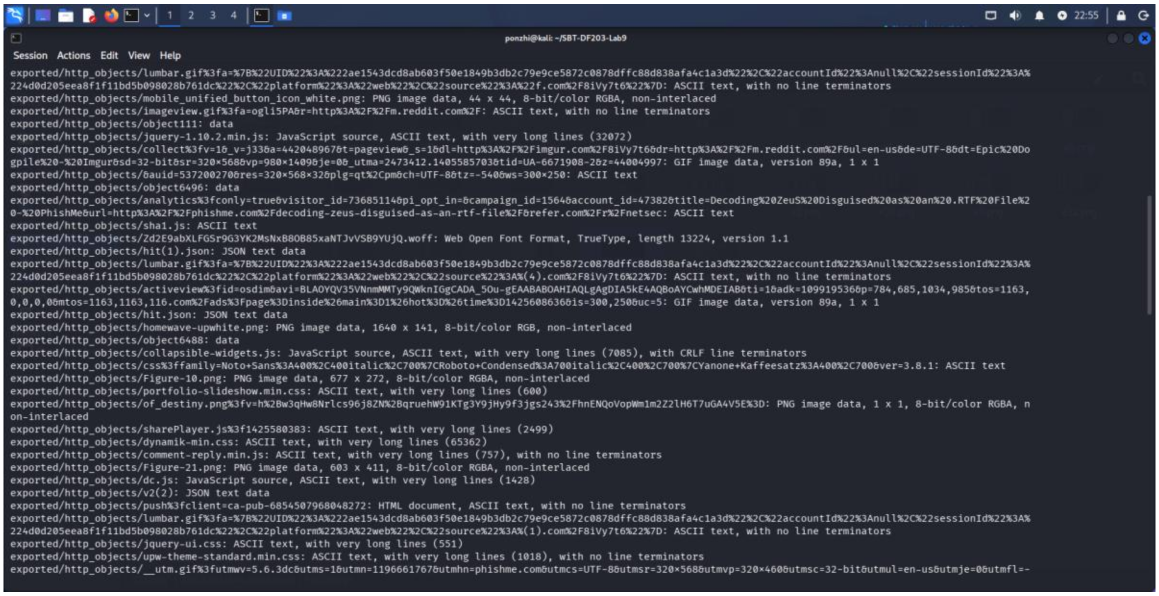
```
mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
```



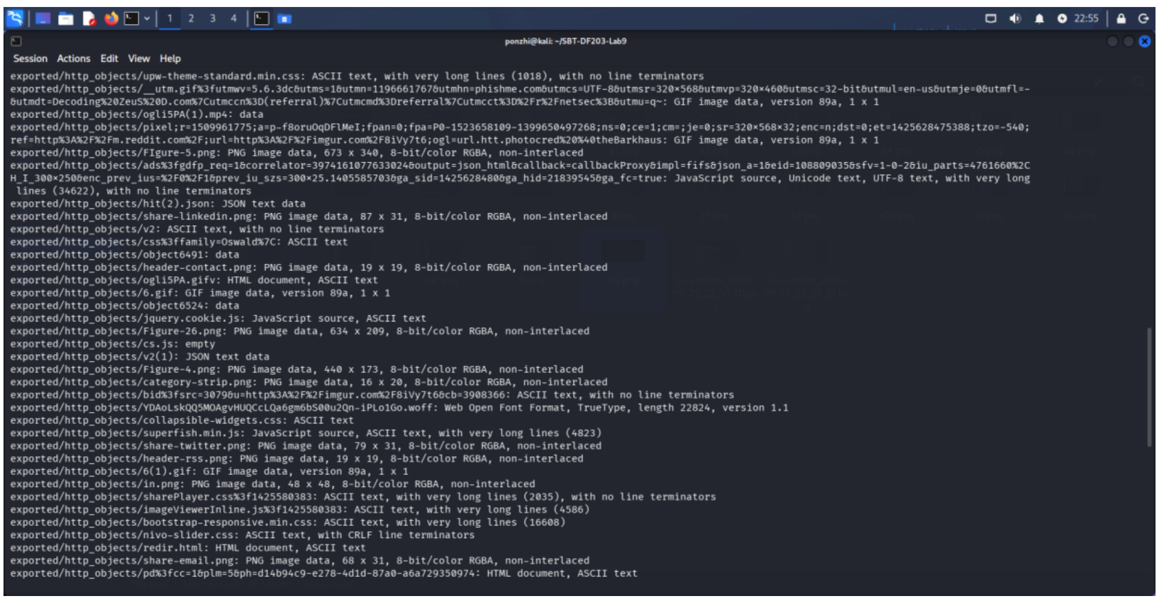
```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help
   27   1.350718      8.8.8.8 → 192.168.0.15 DNS 299 Standard query response 0×be40 A 48-courier.push.apple.com
CNAME 48.courier-push-apple.com.akadns.net CNAME us-courier.push-apple.com.akadns.net A 17.172.239.31 A 17.172.2
39.127 A 17.172.232.188 A 17.172.239.83 A 17.172.232.166 A 17.172.239.25 A 17.172.232.170 A 17.172.232.185
   28   1.350720 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   29   1.351744 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   30   1.353792 192.168.0.15 → 17.172.239.31 TCP 78 50954 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSva
l=549602880 TSecr=0 SACK_PERM
   31   1.556032 192.168.0.15 → 17.172.239.127 TCP 78 50955 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSv
al=549603081 TSecr=0 SACK_PERM
   32   1.703489 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   33   1.703487 17.172.239.31 → 192.168.0.15 TCP 74 443 → 50954 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=1460
 SACK_PERM TSval=646109317 TSecr=549602880 WS=128
   34   1.704513 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   35   1.729089 192.168.0.15 → 17.172.239.31 TCP 66 50954 → 443 [ACK] Seq=1 Ack=1 Win=131760 Len=0 TSval=549603
254 TSecr=646109317
   36   1.743425 192.168.0.15 → 17.172.239.31 TLSv1 255 Client Hello (SNI=courier.push.apple.com)
   37   1.879168 17.172.239.127 → 192.168.0.15 TCP 74 443 → 50955 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=146
0 SACK_PERM TSval=641873548 TSecr=549603081 WS=128
   38   1.906816 192.168.0.15 → 17.172.239.127 TCP 54 50955 → 443 [RST] Seq=1 Win=0 Len=0
   39   1.923199 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   40   1.924226 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   41   2.054781 fe80::1c0c:f6e8:fedb:8f8e → ff02::16     ICMPv6 90 Multicast Listener Report Message v2
   42   2.054784 17.172.239.31 → 192.168.0.15 TCP 66 443 → 50954 [ACK] Seq=1 Ack=190 Win=15616 Len=0 TSval=64610
9707 TSecr=549603268
   43   2.054784 17.172.239.31 → 192.168.0.15 TLSv1 1514 Server Hello
   44   2.054785 17.172.239.31 → 192.168.0.15 TLSv1 1514 Certificate
   45   2.055297 17.172.239.31 → 192.168.0.15 TLSv1 194 Server Key Exchange, Certificate Request, Server Hello D
one
   46   2.056834 fe80::1c0c:f6e8:fedb:8f8e → ff02::16     ICMPv6 90 Multicast Listener Report Message v2
   47   2.088572      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e6be1d
   48   2.092161      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e6be1d
   49   2.094716 192.168.0.15 → 17.172.239.31 TCP 66 50954 → 443 [ACK] Seq=190 Ack=2897 Win=128864 Len=0 TSval=5
49603611 TSecr=646109715
   50   2.094716 192.168.0.15 → 17.172.239.31 TCP 66 50954 → 443 [ACK] Seq=190 Ack=3025 Win=128736 Len=0 TSval=5
49603611 TSecr=646109715
   51   2.095232  192.168.0.1 → 192.168.0.15 DHCP 590 DHCP ACK      - Transaction ID 0×21e6be1d
```

```
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
```

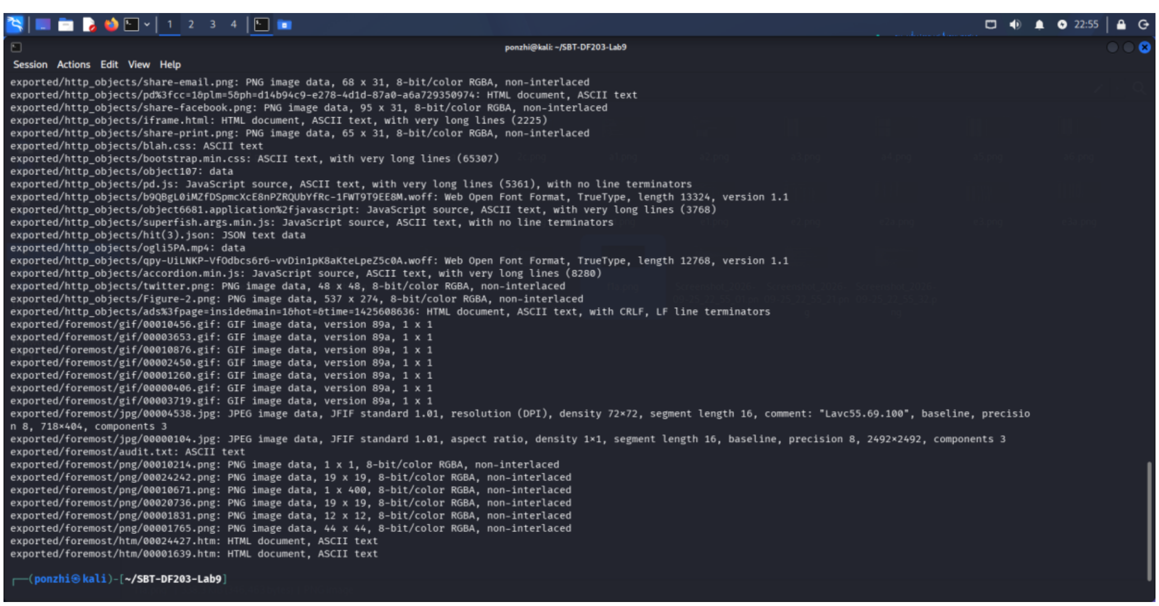
```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
Processing: working/file_working-dec
|*|

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/http_objects/hls%3fversion=4.12: XML 1.0 document, ASCII text
exported/http_objects/object6689.application%2fjavascript: HTML document, ASCII text, with CRLF line terminators
exported/http_objects/0i=19612&llnocookies=undefined: HTML document, ASCII text
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(3).com%2F8iVy7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/Figure-16.png: PNG image data, 230 x 144, 8-bit/color RGBA, non-interlaced
exported/http_objects/analytics%3fver=3&visitor_id=0pi_opt_in=0&campaign_id=1564&account_id=473826title=Decoding%20ZeuS%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&
url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&referrer=http%3A%2.com%2Fr%2Fnetsec: ASCII text
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(2).com%2F8iVy7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/portfolio-slideshow.min.js: JavaScript source, ASCII text, with very long lines (897)
exported/http_objects/rss.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/adc-i-00aecd.png: PNG image data, 12 x 12, 8-bit/color RGBA, non-interlaced
exported/http_objects/rombo.png: PNG image data, 61 x 70, 8-bit/color RGBA, non-interlaced
exported/http_objects/trackalyze.js: ASCII text, with CRLF line terminators
exported/http_objects/dot.gif: GIF image data, version 89a, 1 x 1
exported/http_objects/ogli5PAh.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72×72, segment length 16, comment: "Lavc55.69.100", baseline, precisio
n 8, 718×404, components 3
exported/http_objects/admin-ajax.php%3faction=wordfence_logHumanhid=305E48E6D496430879857CB91BBAF714br=0.1958578866906464: empty
exported/http_objects/page-list.css: ASCII text
exported/http_objects/phone.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/dynamik-responsive.js: JavaScript source, ASCII text
exported/http_objects/logo.png: PNG image data, 194 x 55, 8-bit/color RGBA, non-interlaced
exported/http_objects/tabs.min.js: JavaScript source, ASCII text, with very long lines (11858)
exported/http_objects/%2f: HTML document, ASCII text
exported/http_objects/object6521: data
exported/http_objects/heart.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/ui-bg_glass_75_e6e6e6_1x400.png: PNG image data, 1 x 400, 8-bit/color RGBA, non-interlaced
exported/http_objects/object6684.application%2fjavascript: JavaScript source, ASCII text, with very long lines (6674)
exported/http_objects/FIgure-15.png: PNG image data, 470 x 145, 8-bit/color RGBA, non-interlaced
exported/http_objects/ui-icons_888888_256x240.png: PNG image data, 256 x 240, 8-bit colormap, non-interlaced
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
```



```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22f.com%2F8iVy7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/mobile_unified_button_icon_white.png: PNG image data, 44 x 44, 8-bit/color RGBA, non-interlaced
exported/http_objects/imageview.gif%3fa=ogli5PA&r=http%3A%2F%2Fm.reddit.com%2F: ASCII text, with no line terminators
exported/http_objects/object111: data
exported/http_objects/jquery-1.10.2.min.js: JavaScript source, ASCII text, with very long lines (32072)
exported/http_objects/collect%3fv=1&_v=j33&a=442048967&t=pageview&_s=1&dl=http%3A%2F%2Fimgur.com%2F8iVy7t6&dr=http%3A%2F%2Fm.reddit.com%2F&ul=en-us&de=UTF-8&dt=Epic%20Do
gpile%20-%20Imgur&sd=32-bit&sr=320×568&vp=980×1409&je=0&_utma=2473412.1405585703&tid=UA-6671908-2&z=44004997: GIF image data, version 89a, 1 x 1
exported/http_objects/&auid=537200270&res=320×568×32&plg=qt%2Cpm&ch=UTF-8&tz=-540&ws=300×250: ASCII text
exported/http_objects/object6496: data
exported/http_objects/analytics%3fconly=true&visitor_id=73685114&pi_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20ZeuS%20Disguised%20as%20an%20.RTF%20File%2
0-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2Fr%2Fnetsec: ASCII text
exported/http_objects/sha1.js: ASCII text
exported/http_objects/Zd2E9abXLFGSr9G3YK2MsNxB8OB85xaNTJvVSB9YUjQ.woff: Web Open Font Format, TrueType, length 13224, version 1.1
exported/http_objects/hit(1).json: JSON text data
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(4).com%2F8iVy7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/activeview%3fid=osdim&avi=BLAOYQV35VNnmMMTy9QWknIGgCADA_5Ou-gEAABABOAHIAQLgAgDIA5kE4AQBoAYCwhMDEIAB&ti=1&adk=109919536&p=784,685,1034,985&tos=1163,
0,0,0,0&mtos=1163,1163,116.com%2Fads%3Fpage%3Dinside%26main%3D1%26hot%3D%26time%3D1425608636&is=300,250&uc=5: GIF image data, version 89a, 1 x 1
exported/http_objects/hit.json: JSON text data
exported/http_objects/homewave-upwhite.png: PNG image data, 1640 x 141, 8-bit/color RGB, non-interlaced
exported/http_objects/object6488: data
exported/http_objects/collapsible-widgets.js: JavaScript source, ASCII text, with very long lines (7085), with CRLF line terminators
exported/http_objects/css%3ffamily=Noto+Sans%3A400%2C400italic%2C700%7CRoboto+Condensed%3A700italic%2C400%2C700%7CYanone+Kaffeesatz%3A400%2C700&ver=3.8.1: ASCII text
exported/http_objects/Figure-10.png: PNG image data, 677 x 272, 8-bit/color RGBA, non-interlaced
exported/http_objects/portfolio-slideshow.min.css: ASCII text, with very long lines (600)
exported/http_objects/of_destiny.png%3fv=h%2Bw3qHw8Nrlcs96j8ZN%2BqruehW91KTg3Y9jHy9f3jgs243%2FhnENQoVopWm1m2Z2lH6T7uGA4V5E%3D: PNG image data, 1 x 1, 8-bit/color RGBA, n
on-interlaced
exported/http_objects/sharePlayer.js%3f1425580383: ASCII text, with very long lines (2499)
exported/http_objects/dynamik-min.css: ASCII text, with very long lines (65362)
exported/http_objects/comment-reply.min.js: ASCII text, with very long lines (757), with no line terminators
exported/http_objects/Figure-21.png: PNG image data, 603 x 411, 8-bit/color RGBA, non-interlaced
exported/http_objects/dc.js: JavaScript source, ASCII text, with very long lines (1428)
exported/http_objects/v2(2): JSON text data
exported/http_objects/push%3fclient=ca-pub-6854507968048272: HTML document, ASCII text, with no line terminators
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%
224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(1).com%2F8iVy7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/jquery-ui.css: ASCII text, with very long lines (551)
exported/http_objects/upw-theme-standard.min.css: ASCII text, with very long lines (1018), with no line terminators
exported/http_objects/__utm.gif%3futmwv=5.6.3dc&utms=1&utmn=1196661767&utmhn=phishme.com&utmcs=UTF-8&utmsr=320×568&utmvp=320×460&utmsc=32-bit&utmul=en-us&utmje=0&utmfl=-
```



```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help
exported/http_objects/upw-theme-standard.min.css: ASCII text, with very long lines (1018), with no line terminators
exported/http_objects/__utm.gif%3futmwv=5.6.3dc&utms=1&utmn=1196661767&utmhn=phishme.com&utmcs=UTF-8&utmsr=320×568&utmvp=320×460&utmsc=32-bit&utmul=en-us&utmje=0&utmfl=-
&utmdt=Decoding%20ZeuS%20D.com%7Cutmccn%3D(referral)%7Cutmcmd%3Dreferral%7Cutmcct%3D%2Fr%2Fnetsec%3B&utmu=q~: GIF image data, version 89a, 1 x 1
exported/http_objects/ogli5PA(1).mp4: data
exported/http_objects/pixel;r=1509961775;a=p-f8oruOqDFlMeI;fpan=0;fpa=P0-1523658109-1399650497268;ns=0;ce=1;cm=;je=0;sr=320×568×32;enc=n;dst=0;et=1425628475388;tzo=-540;
ref=http%3A%2F%2Fm.reddit.com%2F;url=http%3A%2F%2Fimgur.com%2F8iVy7t6;ogl=url.htt.photocred%20%40theBarkhaus: GIF image data, version 89a, 1 x 1
exported/http_objects/FIgure-5.png: PNG image data, 673 x 340, 8-bit/color RGBA, non-interlaced
exported/http_objects/ads%3fgdfp_req=1&correlator=3974161077633024&output=json_html&callback=callbackProxy&impl=fifs&json_a=1&eid=108809035&sfv=1-0-2&iu_parts=4761660%2C
H_I_300×250&enc_prev_ius=%2F0%2F1&prev_iu_szs=300×25.1405585703&ga_sid=1425628480&ga_hid=21839545&ga_fc=true: JavaScript source, Unicode text, UTF-8 text, with very long
 lines (34622), with no line terminators
exported/http_objects/hit(2).json: JSON text data
exported/http_objects/share-linkedin.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/v2: ASCII text, with no line terminators
exported/http_objects/css%3ffamily=Oswald%7C: ASCII text
exported/http_objects/object6491: data
exported/http_objects/header-contact.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/ogli5PA.gifv: HTML document, ASCII text
exported/http_objects/6.gif: GIF image data, version 89a, 1 x 1
exported/http_objects/object6524: data
exported/http_objects/jquery.cookie.js: JavaScript source, ASCII text
exported/http_objects/Figure-26.png: PNG image data, 634 x 209, 8-bit/color RGBA, non-interlaced
exported/http_objects/cs.js: empty
exported/http_objects/v2(1): JSON text data
exported/http_objects/Figure-4.png: PNG image data, 440 x 173, 8-bit/color RGBA, non-interlaced
exported/http_objects/category-strip.png: PNG image data, 16 x 20, 8-bit/color RGBA, non-interlaced
exported/http_objects/bid%3fsrc=3079&u=http%3A%2F%2Fimgur.com%2F8iVy7t6&cb=3908366: ASCII text, with no line terminators
exported/http_objects/YDAoLskQQ5MOAgvHUQCcLQa6gm6bS00u2Qn-iPLo1Go.woff: Web Open Font Format, TrueType, length 22824, version 1.1
exported/http_objects/collapsible-widgets.css: ASCII text
exported/http_objects/superfish.min.js: JavaScript source, ASCII text, with very long lines (4823)
exported/http_objects/share-twitter.png: PNG image data, 79 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/header-rss.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/6(1).gif: GIF image data, version 89a, 1 x 1
exported/http_objects/in.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/sharePlayer.css%3f1425580383: ASCII text, with very long lines (2035), with no line terminators
exported/http_objects/imageViewerInline.js%3f1425580383: ASCII text, with very long lines (4586)
exported/http_objects/bootstrap-responsive.min.css: ASCII text, with very long lines (16608)
exported/http_objects/nivo-slider.css: ASCII text, with CRLF line terminators
exported/http_objects/redir.html: HTML document, ASCII text
exported/http_objects/share-email.png: PNG image data, 68 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/pd%3fcc=1&plm=5&ph=d14b94c9-e278-4d1d-87a0-a6a729350974: HTML document, ASCII text
```



```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help
exported/http_objects/share-email.png: PNG image data, 68 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/pd%3fcc=1&plm=5&ph=d14b94c9-e278-4d1d-87a0-a6a729350974: HTML document, ASCII text
exported/http_objects/share-facebook.png: PNG image data, 95 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/iframe.html: HTML document, ASCII text, with very long lines (2225)
exported/http_objects/share-print.png: PNG image data, 65 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/blah.css: ASCII text
exported/http_objects/bootstrap.min.css: ASCII text, with very long lines (65307)
exported/http_objects/object107: data
exported/http_objects/pd.js: JavaScript source, ASCII text, with very long lines (5361), with no line terminators
exported/http_objects/b9QBgL0iMZfDSpmcXcE8nPZRQUbYfRc-1FWT9T9EE8M.woff: Web Open Font Format, TrueType, length 13324, version 1.1
exported/http_objects/object6681.application%2fjavascript: JavaScript source, ASCII text, with very long lines (3768)
exported/http_objects/superfish.args.min.js: JavaScript source, ASCII text, with no line terminators
exported/http_objects/hit(3).json: JSON text data
exported/http_objects/ogli5PA.mp4: data
exported/http_objects/qpy-UiLNKP-VfOdbcs6r6-vvDin1pK8aKteLpeZ5c0A.woff: Web Open Font Format, TrueType, length 12768, version 1.1
exported/http_objects/accordion.min.js: JavaScript source, ASCII text, with very long lines (8280)
exported/http_objects/twitter.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/Figure-2.png: PNG image data, 537 x 274, 8-bit/color RGBA, non-interlaced
exported/http_objects/ads%3fpage=inside&main=1&hot=&time=1425608636: HTML document, ASCII text, with CRLF, LF line terminators
exported/foremost/gif/00010456.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010876.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00002450.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003719.gif: GIF image data, version 89a, 1 x 1
exported/foremost/jpg/00004538.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72×72, segment length 16, comment: "Lavc55.69.100", baseline, precisio
n 8, 718×404, components 3
exported/foremost/jpg/00000104.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1×1, segment length 16, baseline, precision 8, 2492×2492, components 3
exported/foremost/audit.txt: ASCII text
exported/foremost/png/00010214.png: PNG image data, 1 x 1, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00024242.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00010671.png: PNG image data, 1 x 400, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00020736.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00001831.png: PNG image data, 12 x 12, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00001765.png: PNG image data, 44 x 44, 8-bit/color RGBA, non-interlaced
exported/foremost/htm/00024427.htm: HTML document, ASCII text
exported/foremost/htm/00001639.htm: HTML document, ASCII text

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
```

find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help

┌──(ponzhi㉿kali)-[~/SBT-DF203-Lab9]
└─$ find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
0868db67f79acd281fcb3b46c9768dae7033be91f9551185394abf19d4b7997ae  exported/http_objects/hls%3fversion=4.12
9936530c4778e5cdda0028673cab8f3ad0cf9cf589123a6e9d15413e1fd1a083  exported/http_objects/object6689.application%2fjavascript
c7d473663722fe9d8d2587d1f1d3ef2603af41860d6cdadbeac32705e3f0c8b7  exported/http_objects/bi=196120linocookies=undefined
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b  exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%2
2%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(3).com%2F8iVy7t6%22%7D
2f97328c08c8b6e8279e4cba83b95f1faf4e264a07e593c55b3e6753f5d1a0c6  exported/http_objects/Figure-16.png
0c32950b6a816d5053e1b495c6b148bcfa4260d7cca890f6f95ba310ca3490db  exported/http_objects/analytics%3fver=3&visitor_id=6pi_opt_in=&campaign_id=1564&account_id=473820title=Decoding%20Zeu5%20
Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&referrer=http%3A%2.com%2Fr%2Fnetsec
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b  exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%2
2%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(2).com%2F8iVy7t6%22%7D
5d42025c1f6793c70cc31512f2d71b9bd8466d02609fdfc46882c02d3726f21a  exported/http_objects/portfolio-slideshow.min.js
543ff93cef2ec75b4f3e4ca0cda282fbf0ad18f55201eb3d295f738ac818cbea  exported/http_objects/rss.png
534c00b9b2d436a30c260eaad82679d2a2c970d4962399a8da227ee07162a755  exported/http_objects/adc-i-00aecd.png
eb5c89dca6a23af103550d1c749ef0027b44a644bc371e3a560e0054495928c73  exported/http_objects/rombo.png
59ebace7202b39be46546a909c5ae92c2a34245cba32537b63564c22169e34cf  exported/http_objects/trackalyze.js
53f7e56783c414746a9fc3612637509560424617e31352486 4eac142d6e0f94  exported/http_objects/dot.gif
d89b41b407c4af80ec8901c0ab246e3aafde949da0bbb30d95f1d641cb7c87a8  exported/http_objects/ogli5PAh.jpg
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855  exported/http_objects/admin-ajax.php%3faction=wordfence_logHumanbhid=305E48E6D496430879857CB91BBAF7146r=0.195857886690646
4
59ebc47331c997b569e83c98089d44b1f3797e2bec0e6cf979f8c7e43643f452  exported/http_objects/page-list.css
1a9789f6a752a3646433ea6c1b78c78405982e6970934ee26e99396e48ddb001  exported/http_objects/phone.png
b500fbbfef27ffcd1573ccf334748e6969e3f9f8ab463653ec23951fd5093cb6  exported/http_objects/dynamik-responsive.js
4b330bc7f2e4faab709f44958cb276cfb497ede39138aa0e8a4fbf4fc5a69653  exported/http_objects/logo.png
cbee98abc14fa8f2220de7d00fe9b90bbba64e9344ce66d092da9b08f822e494  exported/http_objects/tabs.min.js
02e3572ea1b28a49e6a710885744310197a0a96d99bb18916b1cbc547ea79e  exported/http_objects/%2f
811876a5865b50b15860070d4c6aabc82967b6bfe8519eb16a844d3e94523a6c  exported/http_objects/object6521
b3dfb88bb3ab92c1dedfb8a54c83606eb922bd2914c5500af9ea66f78e251e3a  exported/http_objects/heart.png
ddf5dd4e0ef2b185e8bb0af7b6e90ebe74a84384cb4700658e76e754c8bfe550  exported/http_objects/ui-bg_glass_75_e6e6e6_1x400.png
95e43f1f5fbd8289bd5aa0de6bb39e33c5d1b8324c825b12a3c1433d93322268  exported/http_objects/object6684.application%2fjavascript
77059c4edefe07cc4d03b5d3d7513b07494bf386f0063369 15e956c8a8c7feaf  exported/http_objects/FIgure-15.png
ea2e29625de3463465e93b002b065f5833e05b97f7a052b1c141e754d62e1a8b  exported/http_objects/ui-icons_888888_256x240.png
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b  exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dffc88d838afa4c1a3d%2
2%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22f.com%2F8iVy7t6%22%7D
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d  exported/http_objects/mobile_unified_button_icon_white.png
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b  exported/http_objects/imageview.gif%3fa=ogli5PA&r=http%3A%2F%2Fm.reddit.com%2F
bc519bf14a7e4effda93b8d2f9eff4aa3753f69dd4f563f561f95341eeb6a8b3  exported/http_objects/object111
0ba081f546084bd5097aa8a73c75931d5aa1fc4d6e846e53c21f98e6a1509988  exported/http_objects/jquery-1.10.2.min.js
```

```
ponzhi@kali: ~/SBT-DF203-Lab9
Session  Actions  Edit  View  Help
bc519bf14a7e4effda93b8d2f9eff4aa3753f69dd4f563f561f95341eeb6a8b3  exported/http_objects/object111
0ba081f546084bd5097aa8a73c75931d5aa1fc4d6e846e53c21f98e6a1509988  exported/http_objects/jquery-1.10.2.min.js
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/http_objects/collect%3fv=1&_v=j33&a=442048967&t=pageview&_s=1&dl=http%3A%2F%2Fimgur.com%2F8iVy7t6&dr=http%3A%2F%
2Fm.reddit.com%2F&ul=en-us&de=UTF-8&dt=Epic%20Dogpile%20-%20Imgur&sd=32-bit&sr=320x568&vp=980x14096je=0&_utma=2473412.1405585703&tid=UA-6671908-2&z=44004997
187f2ad17d1fee50d8625e4be5b02d35c7f6cde4209d458a3717f739a25860e4  exported/http_objects/&auid=537200270&res=320x568x32&plg=qt%2Cpm&ch=UTF-8&tz=-540&ws=300x250
974bd921fe2d8a0b997d9f8bc12a676ef49920663be48813fe6b0b15e65bc8fb  exported/http_objects/object6496
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ing%20Zeu5%20Disguised%20as%20an%20.RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2Fr%2Fnetsec
44faa8e2ed83768b8c23763c4a4fc5856d9790be44913a77c5a2ad1675ba3f24  exported/http_objects/sha1.js
5eafcd482cbd31c604c630b54ce55bfc87ce483f25a9b91a8c2c2fd4c4557d15  exported/http_objects/Zd2E9abXLFGSr9G3YK2MsNxB8OB85xaNTJvVSB9YUjQ.woff
902a97446676157ac302761ad4bf7241655853ec46f8b4e3d0db87023b900b4  exported/http_objects/hit(1).json
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2%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%(4).com%2F8iVy7t6%22%7D
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```

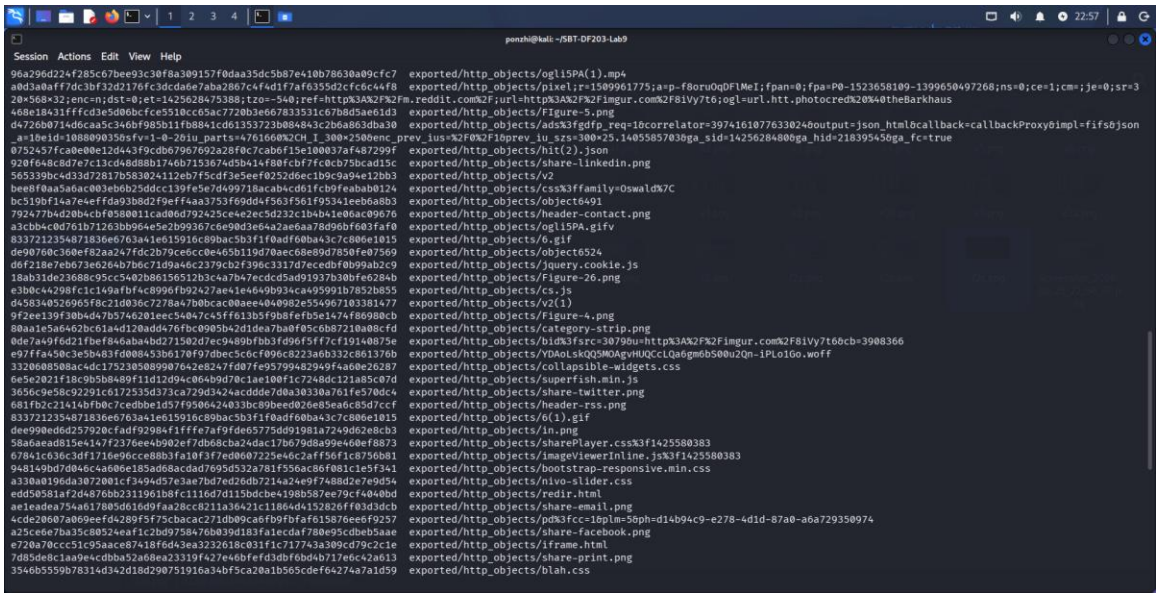


# Required Findings Worksheet

| Question | Finding |
|---|---|
| Capture format and duration | IEEE 802.11 pcap, 45,169 packets, 272.97 s (2015-03-06 08:51:54.364525 – 08:56:27.334320) |
| BSSID | 00:26:66:55:97:d6 (ESSID: cgnetwork) |
| Primary station MAC addresses | Host: f0:f6:1c:68:96:7c (Apple_68:96:7c); Gateway/AP: 00:26:66:55:97:d4 (EFMNetworks_55:97:d4); secondary station: ASUSTekCOMPU_2a:c2:7a (10 stations total per airdecap-ng) |
| Protected frame count | 15,477 WEP data frames |
| Repeated IV evidence | Multiple IVs repeat 4–7 times each (e.g. 0x42e8eb ×7, 0x28d3eb ×7) — confirms IV-space exhaustion/reuse |
| Recovered/validated WEP40 key | A4:3D:F6:F3:74 (Aircrack-ng PTW attack; matches slide-provided key; 100% decryption success) |
| Decrypted capture filename/hash | working/file_working-dec — SHA-256 167c91994c269777f9048227deb89882caf3fc3c763977f2059604f9a6a40b04 |
| Host IP and MAC | 192.168.0.15 / f0:f6:1c:68:96:7c (Apple_68:96:7c) |
| Server IP and MAC | No remote-host MAC is observable (traffic is forwarded via gateway 00:26:66:55:97:d4); heaviest remote IPs are 198.90.20.111 and 199.27.79.193, with HTTP content also served from 198.41.209.137 and 23.21.196.154 |
| Key protocols observed | ARP, DHCP, DNS, ICMPv6, IGMPv2/MDNS, TCP, TLSv1/TLSv1.2, HTTP |
| Recovered images/HTML | Dozens of HTTP objects recovered (images, JS, CSS, fonts, JSON, HTML) tied to a PhishMe.com article “Decoding ZeuS Disguised as an RTF File”, plus a supplementary foremost carve of smaller image/HTML fragments — see Section 13 tables and reports/exported_file_hashes.txt for the full hash list |
| Optional router passphrase | Protected appendix only — Part G not performed in this session |

| Question | Finding |
| --- | --- |
| Security conclusion | WEP40's 24-bit IV space and RC4 keystream reuse make key recovery practical with commodity tools (Aircrack-ng) against a modest packet count; once broken, WEP provides no confidentiality at all. WEP should be retired in favor of WPA2-AES/CCMP or WPA3. |

## Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

**Trainee: Kumpon Emmanuel**
Date: 25 September 2026